

Biophysics (Bio215)

TA:

Bhumika Choudhary (bhumikac@iiitd.ac.in)

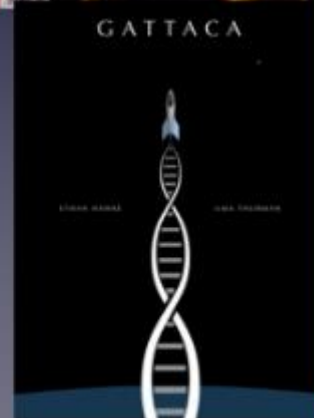
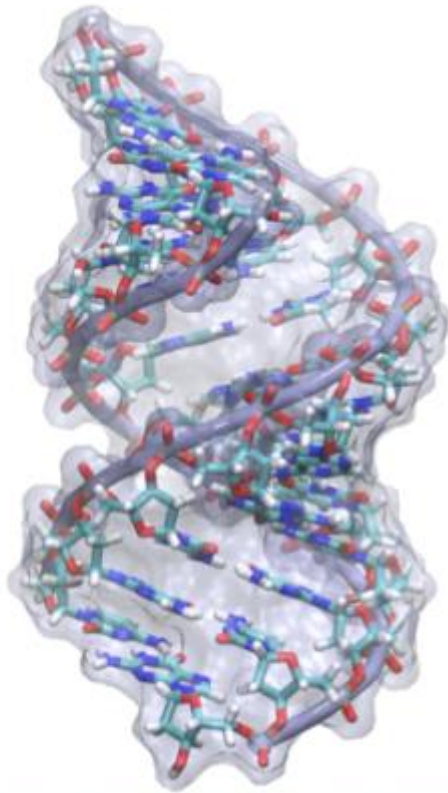
Deeksha Yadav (deekshay@iiitd.ac.in)

Dr. Arjun Ray
(arjun@iiitd.ac.in)

Today's lecture

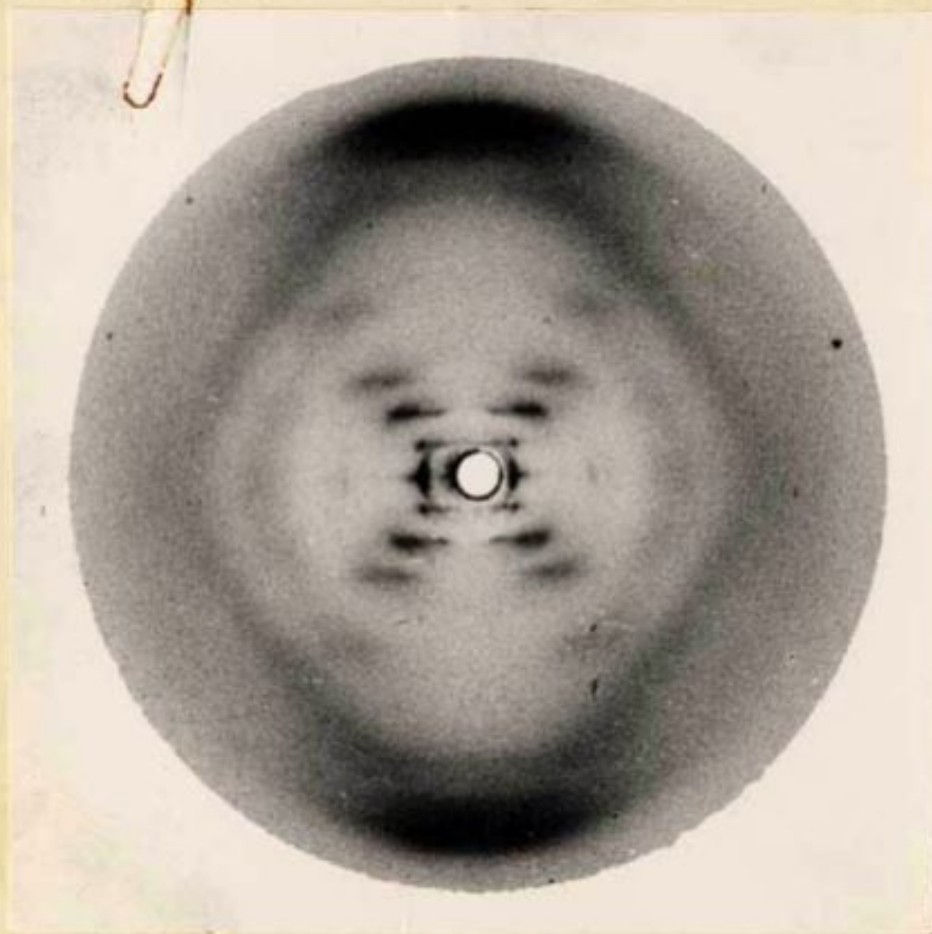
- DNA & RNA
- central dogma of molecular biology
- X-ray crystallography
- Cryo-electron microscopy (cryo-EM)
- Protein structure & function
- Water & Hydrogen bonds

DNA - the molecule of life





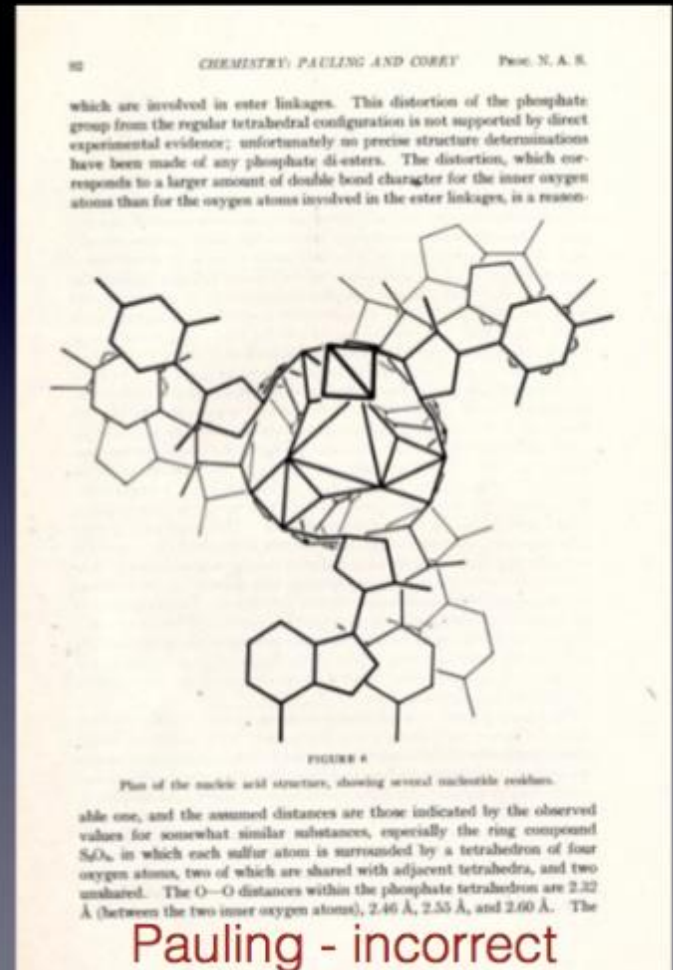
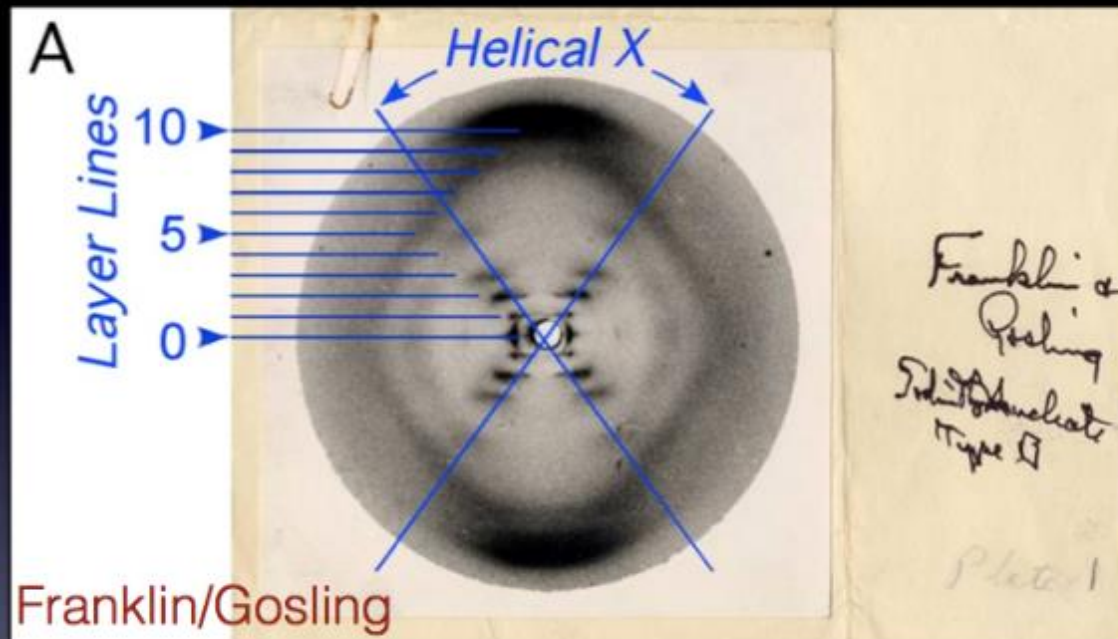
Rosalind Franklin

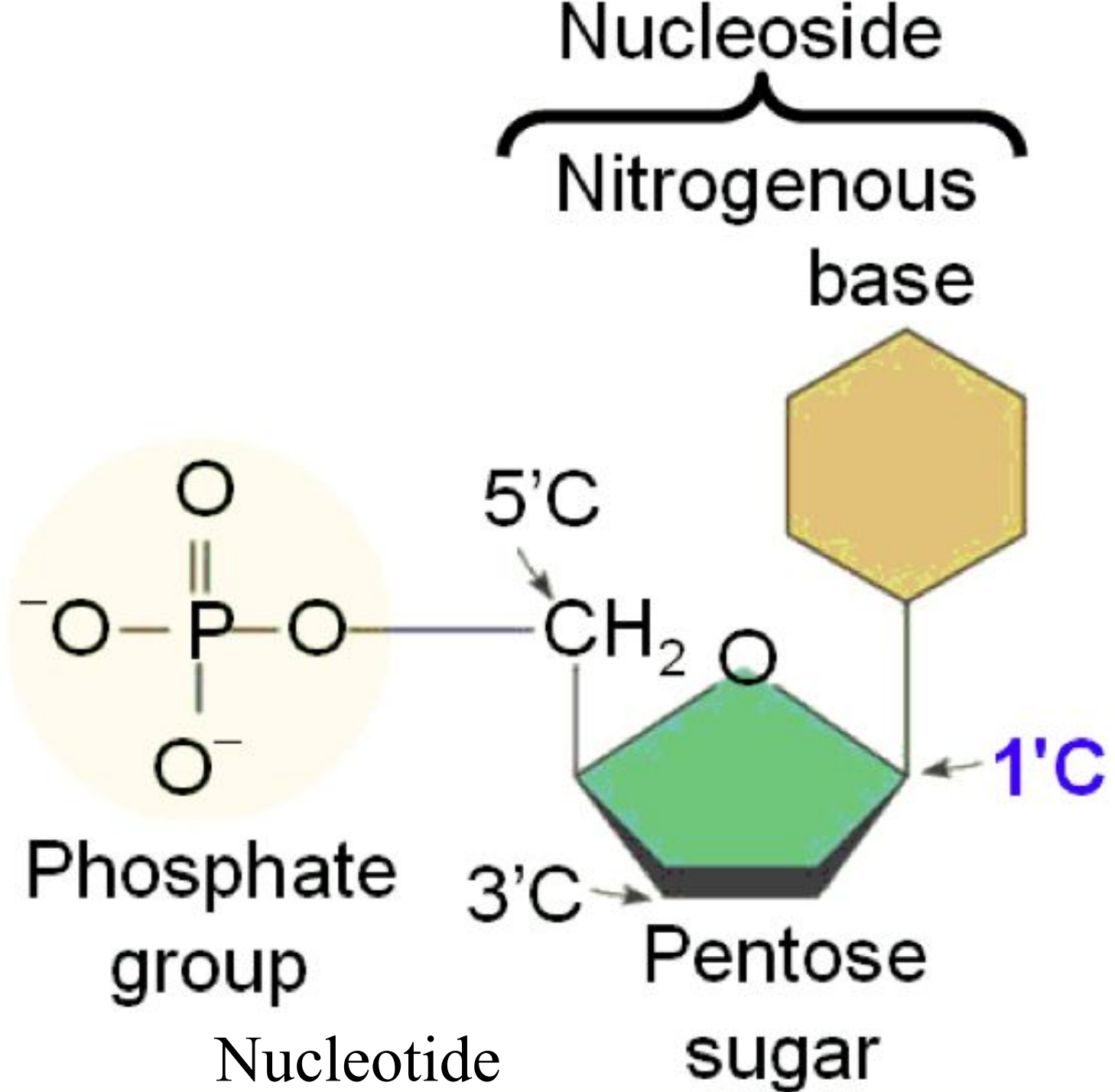


Franklin &
Gosling
5th March 1953
Type D

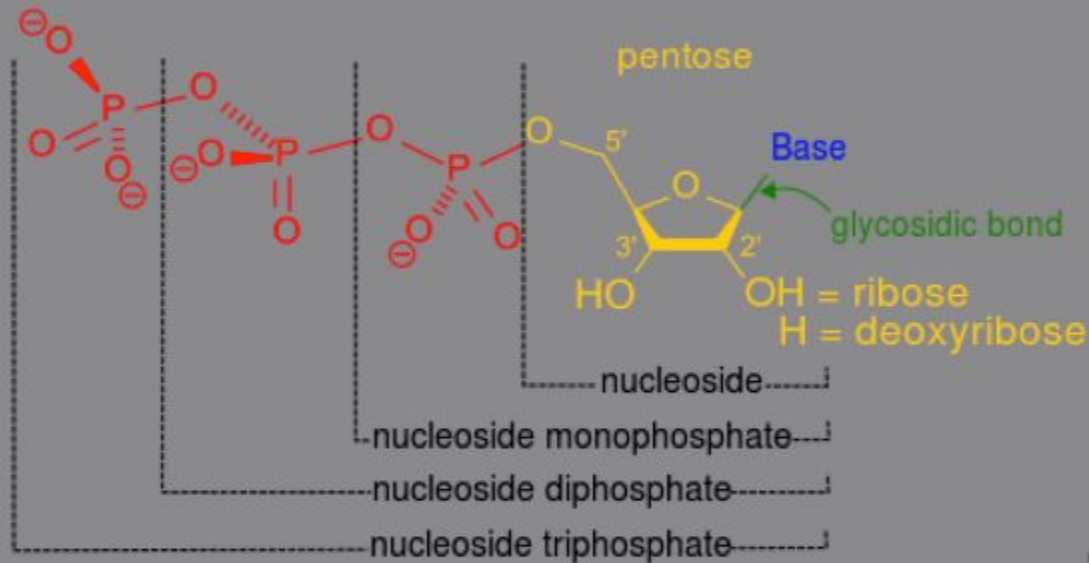
Plate 1

The structure of DNA

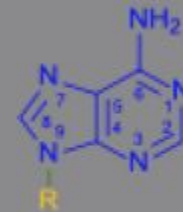




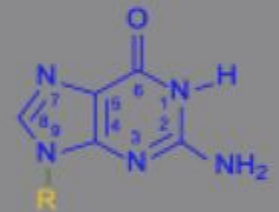
Structure of nucleic acids



Purines

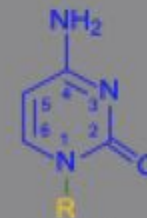


Adenine

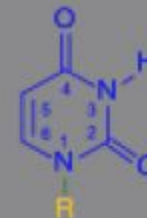


Guanine

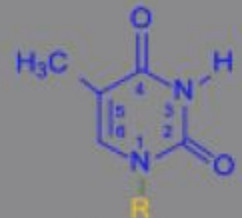
Pyrimidines



Cytosine



Uracil



Thymine

Chargaff's ratios

Table 11-1. Molar Properties of Bases* in DNAs from Various Sources

Organism	Tissue	Adenine	Thymine	Guanine	Cytosine
<i>Escherichia coli</i> (K12)	—	26.0	23.9	24.9	25.2
<i>Diplococcus pneumoniae</i>	—	29.8	31.6	20.5	18.0
<i>Mycobacterium tuberculosis</i>	—	15.1	14.6	34.9	35.4
Yeast	—	31.3	32.9	18.7	17.1
<i>Paracentrotus lividus</i> (sea urchin)	Sperm	32.8	32.1	17.7	18.4
Herring	Sperm	27.8	27.5	22.2	22.6
Rat	Bone marrow	28.6	28.4	21.4	21.5
Human	Thymus	30.9	29.4	19.9	19.8
Human	Liver	30.3	30.3	19.5	19.9
Human	Sperm	30.7	31.2	19.3	18.8

* Defined as moles of nitrogenous constituents per 100 g-atoms phosphate in hydrolysate.

SOURCE: E. Chargaff and J. Davidson, eds., *The Nucleic Acids*. Academic Press, 1955.

which are involved in ester linkages. This distortion of the phosphate group from the regular tetrahedral configuration is not supported by direct experimental evidence; unfortunately no precise structure determinations have been made of any phosphate di-esters. The distortion, which corresponds to a larger amount of double bond character for the inner oxygen atoms than for the oxygen atoms involved in the ester linkages, is a reason-

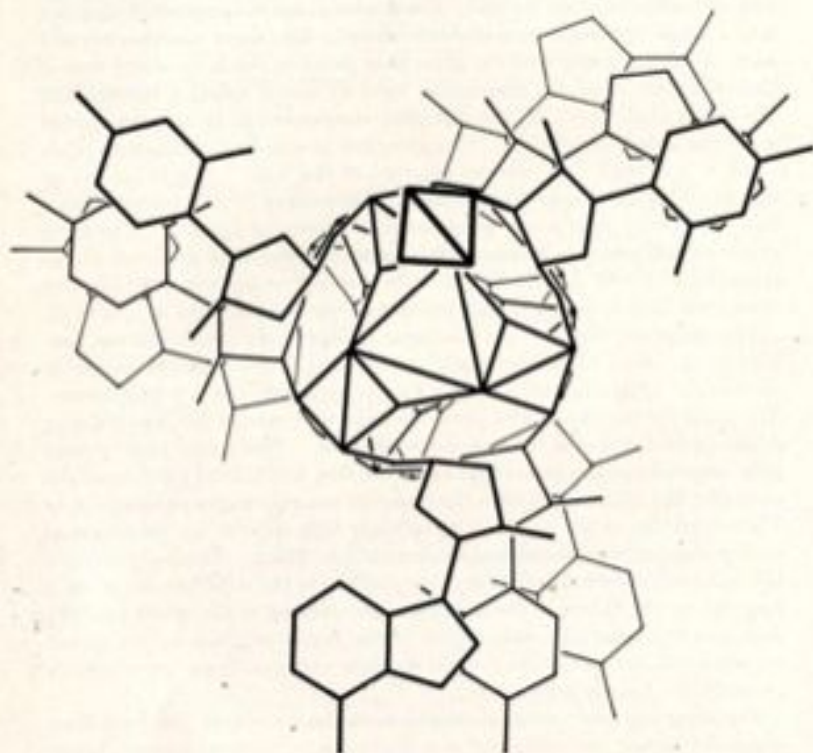


FIGURE 6

Plan of the nucleic acid structure, showing several nucleotide residues.

able one, and the assumed distances are those indicated by the observed values for somewhat similar substances, especially the ring compound S_4O_6 , in which each sulfur atom is surrounded by a tetrahedron of four oxygen atoms, two of which are shared with adjacent tetrahedra, and two unshared. The O—O distances within the phosphate tetrahedron are 2.32 Å (between the two inner oxygen atoms), 2.46 Å, 2.55 Å, and 2.60 Å. The

What makes this structure improbable?

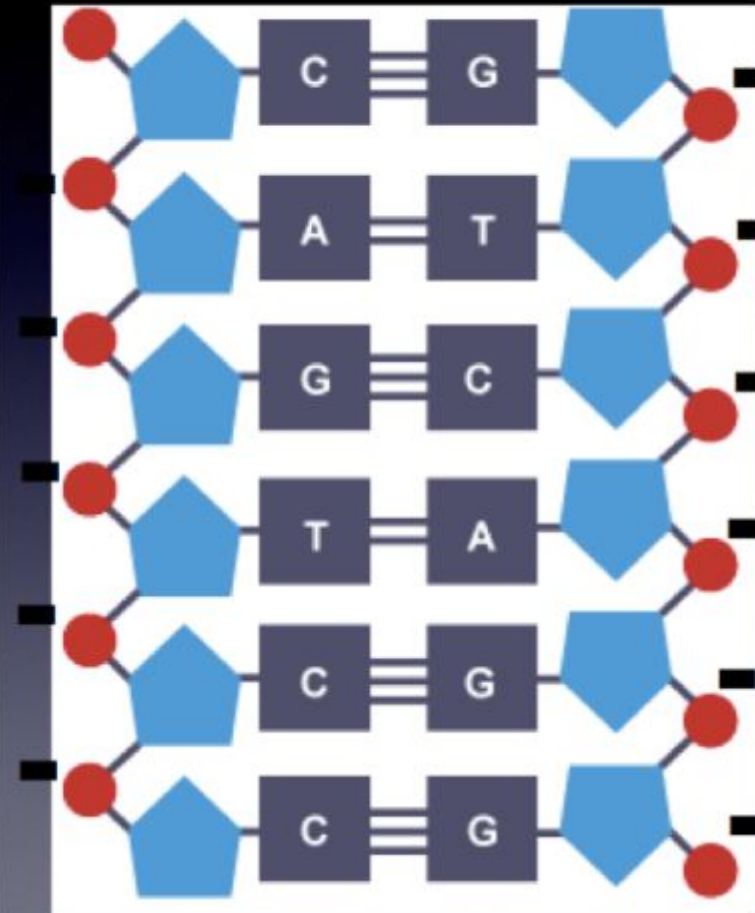
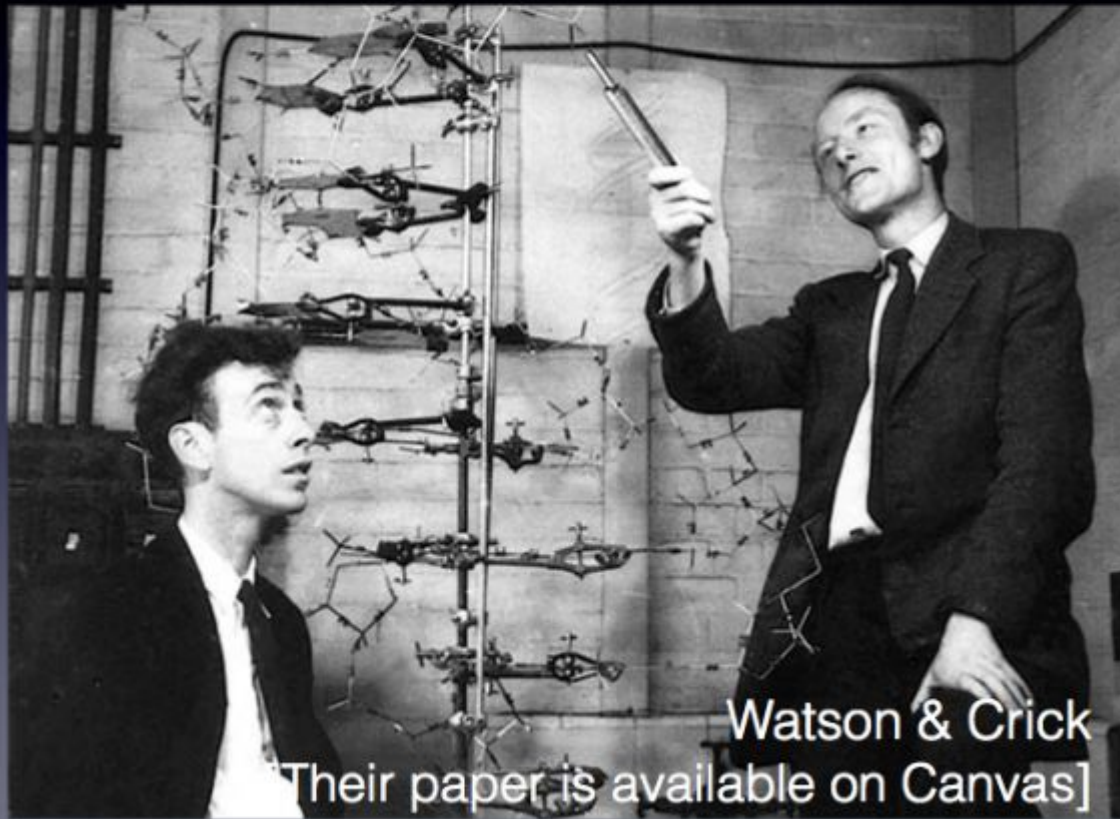
A.) outward-facing bases

B.) repelling electrostatic charge

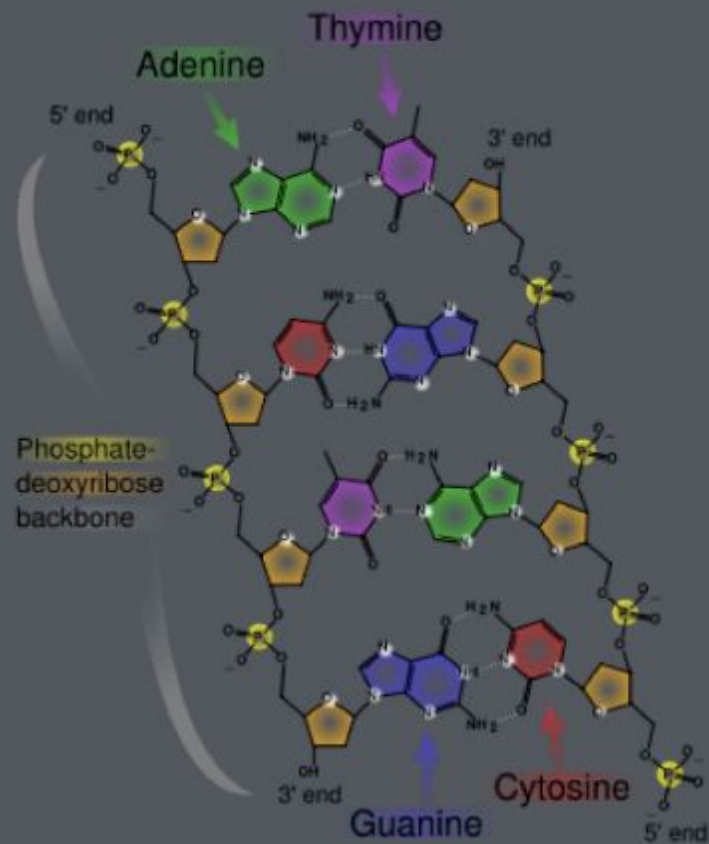
C.) steric clashes

Pauling & Corey, 1953

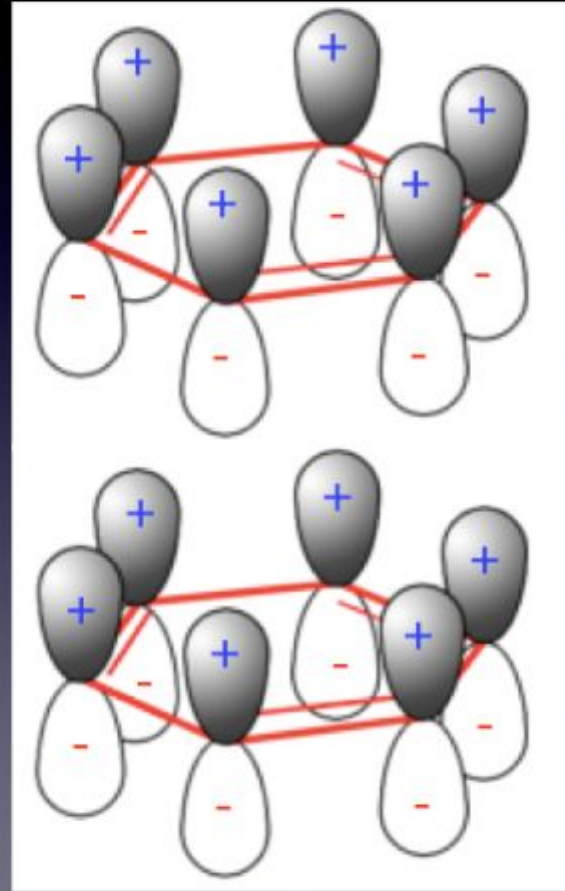
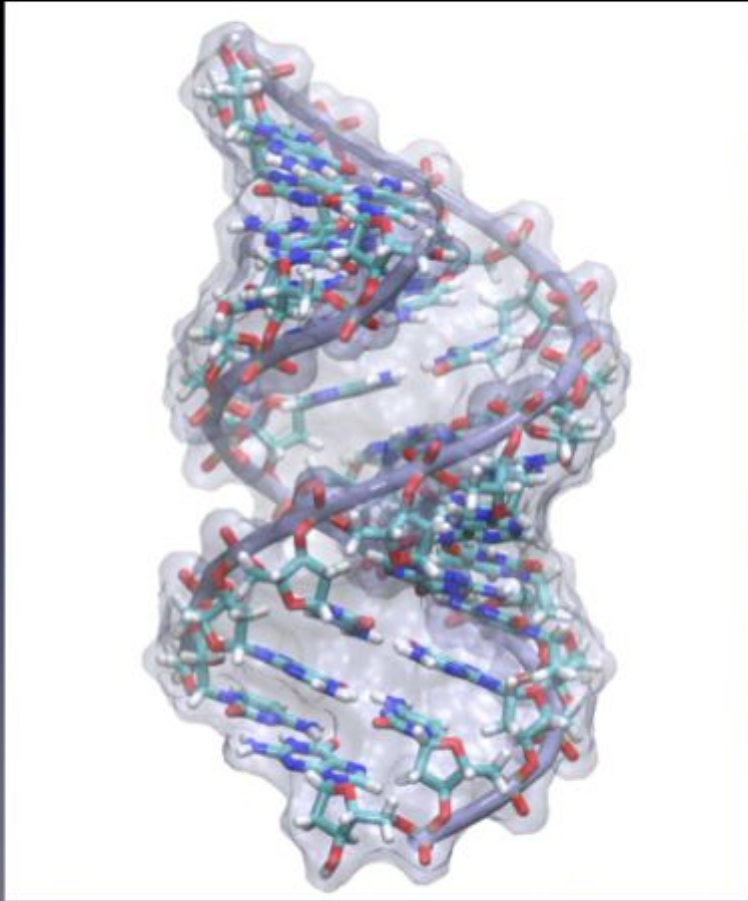
The double helix



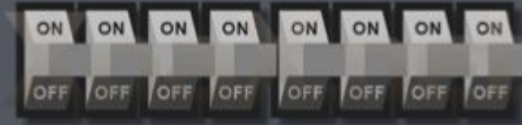
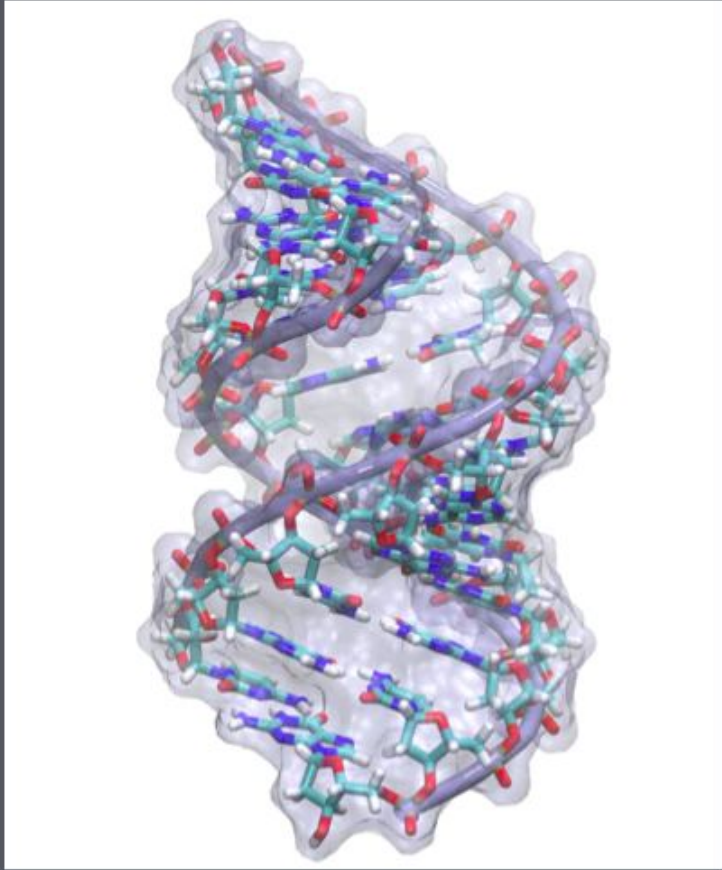
Stabilisation



Stabilisation II



DNA function: Simplicity vs Complexity



$$2^8 = 256$$

$$4^8 = 65,536$$

An integrated encyclopedia of DNA elements in the human genome

The ENCODE Project Consortium*

The human genome encodes the blueprint of life, but the function of the vast majority of its nearly three billion bases is unknown. The Encyclopedia of DNA Elements (ENCODE) project has systematically mapped regions of transcription, transcription factor association, chromatin structure and histone modification. **These data enabled us to assign biochemical functions for 80% of the genome**, in particular outside of the well-studied protein-coding regions. Many discovered candidate regulatory elements are physically associated with one another and with expressed genes, providing new insights into the mechanisms of gene regulation. The newly identified elements also show a statistical correspondence to sequence variants linked to human disease, and can thereby guide interpretation of this variation. Overall, the project provides new insights into the organization and regulation of our genes and genome, and is an expansive resource of functional annotations for biomedical research.

The human genome sequence provides the underlying code for human biology. Despite intensive study, especially in identifying



ENCODE

Encyclopedia of DNA Elements

95% of the genome lies within 8 kilobases (kb) of a DNA–protein interaction (as assayed by bound ChIP-seq motifs or DNase I footprints),



8.2% of the Human Genome Is Constrained: Variation in Rates of Turnover across Functional Element Classes in the Human Lineage

Chris M. Rands¹, Stephen Meader¹, Chris P. Ponting^{1*}, Gerton Lunter^{2*}

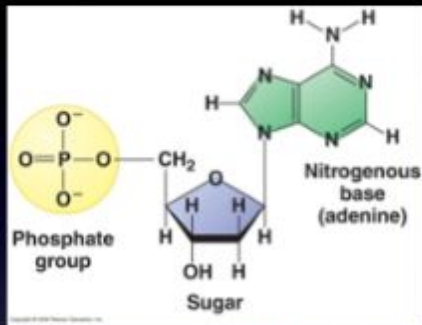
1 MRC Functional Genomics Unit, Department of Physiology, Anatomy, and Genetics, University of Oxford, Oxford, United Kingdom, **2** Wellcome Trust Centre for Human Genetics, University of Oxford, Oxford, United Kingdom

Abstract

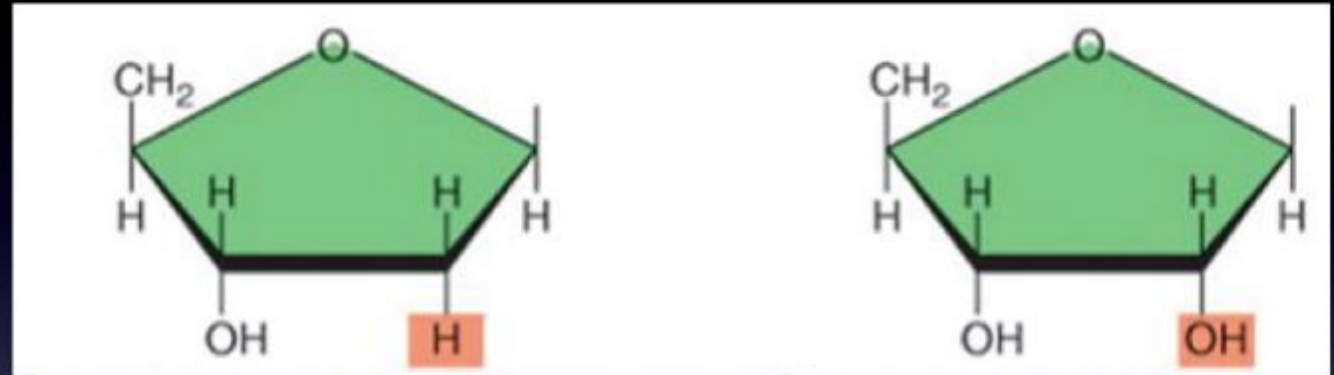
Ten years on from the finishing of the human reference genome sequence, it remains unclear what fraction of the human genome confers function, where this sequence resides, and how much is shared with other mammalian species. When addressing these questions, functional sequence has often been equated with pan-mammalian conserved sequence. However, functional elements that are short-lived, including those contributing to species-specific biology, will not leave a footprint of long-lasting negative selection. Here, we address these issues by identifying and characterising sequence that has been constrained with respect to insertions and deletions for pairs of eutherian genomes over a range of divergences. Within noncoding sequence, we find increasing amounts of mutually constrained sequence as species pairs become more closely related, indicating that noncoding constrained sequence turns over rapidly. We estimate that half of present-day noncoding constrained sequence has been gained or lost in approximately the last 130 million years (half-life in units of divergence time, $d_{1/2}=0.25-0.31$). While enriched with ENCODE biochemical annotations, much of the short-lived constrained sequences we identify are not detected by models optimized for wider pan-mammalian conservation. Constrained DNase 1 hypersensitivity sites, promoters and untranslated regions have been more evolutionarily stable than long noncoding RNA loci which have turned over especially rapidly. By contrast, protein coding sequence has been highly stable, with an estimated half-life of over a billion years ($d_{1/2}=2.1-5.0$). From extrapolations we estimate that 8.2% (7.1–9.2%) of the human genome is presently subject to negative selection and thus is likely to be functional, while only 2.2% has maintained constraint in both human and mouse since these species diverged. These results reveal that the evolutionary history of the human genome has been highly dynamic, particularly for its noncoding yet biologically functional fraction.

Citation: Rands CM, Meader S, Ponting CP, Lunter G (2014) 8.2% of the Human Genome Is Constrained: Variation in Rates of Turnover across Functional Element Classes in the Human Lineage. PLoS Genet 10(7): e1004525. doi:10.1371/journal.pgen.1004525

RNA = RiboNucleicAcid

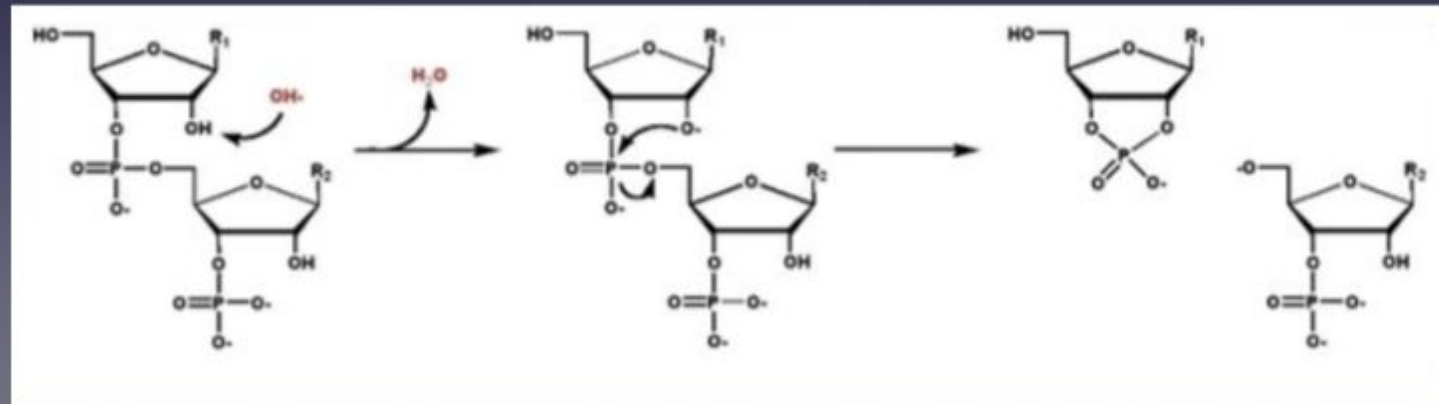


DNA nucleotide



deoxyribose (in DNA)

ribose (in RNA)



Structures

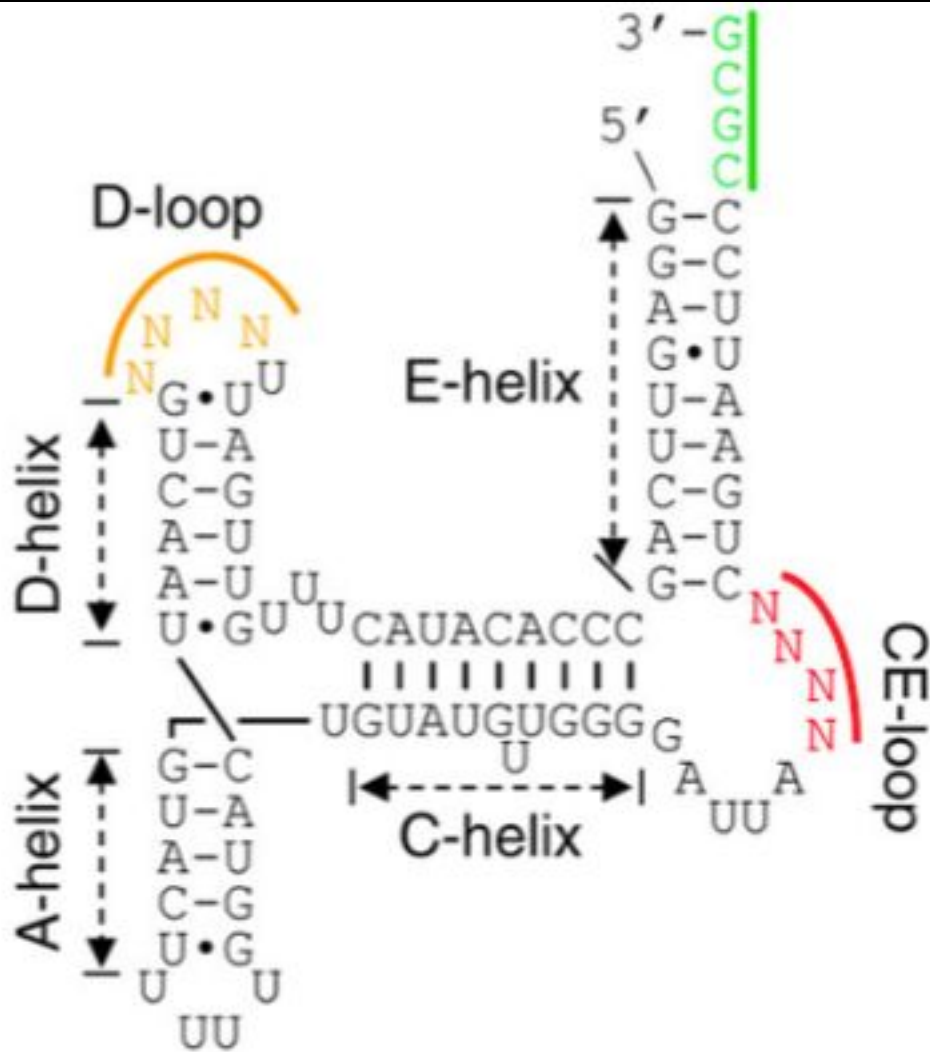


DNA
Deoxyribonucleic acid

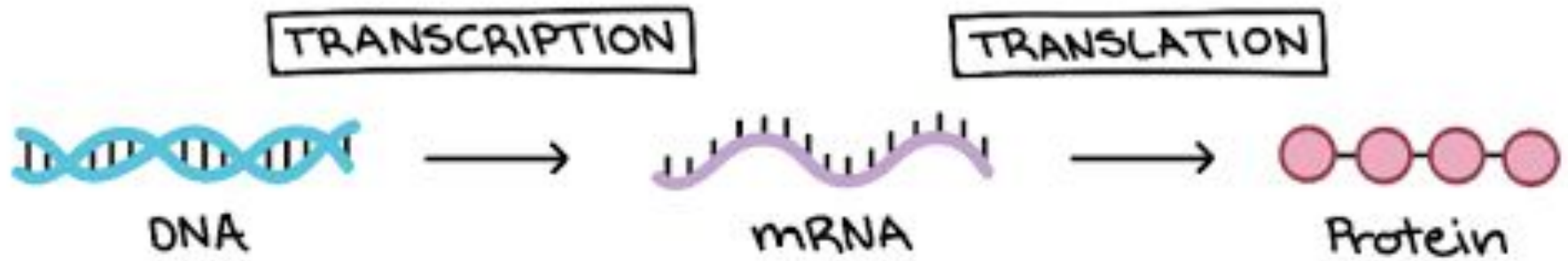


RNA
Ribonucleic acid

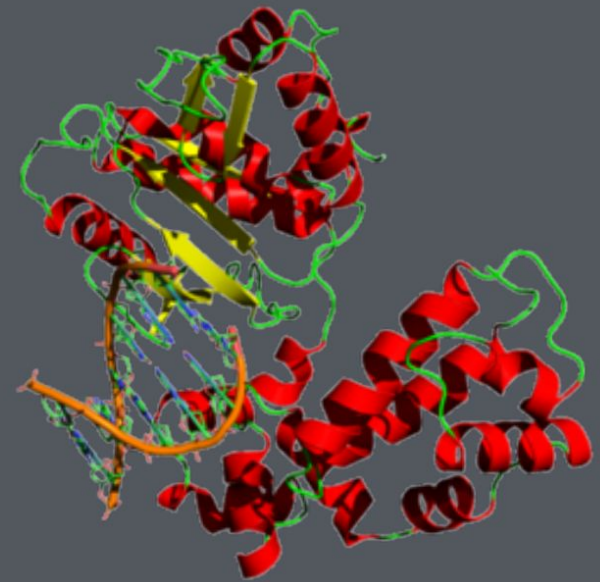
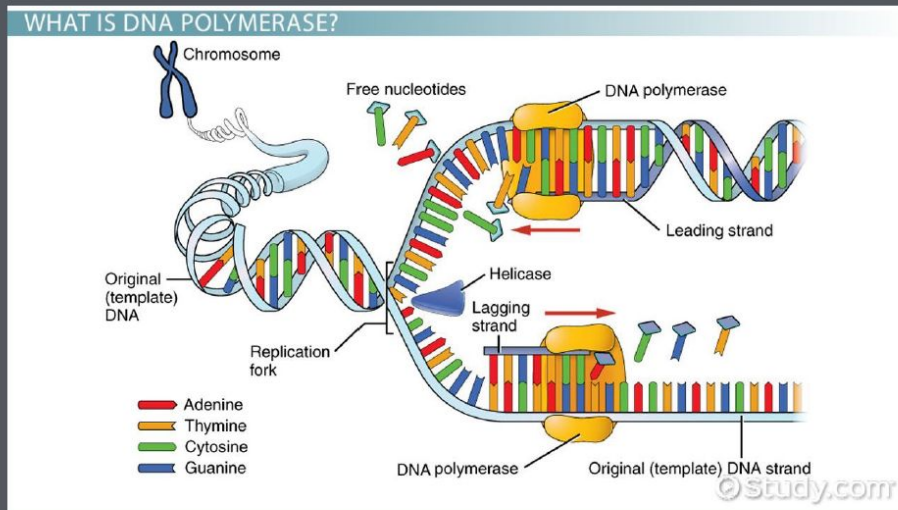
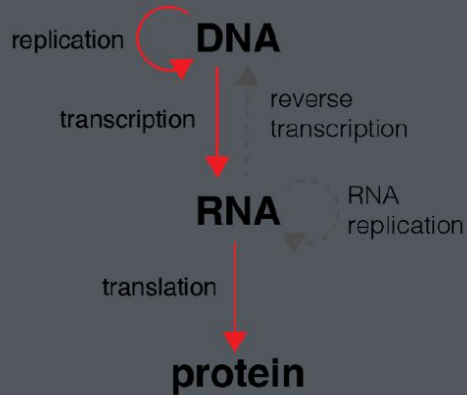
Self-Complimentarity



Central Dogma of Molecular Biology

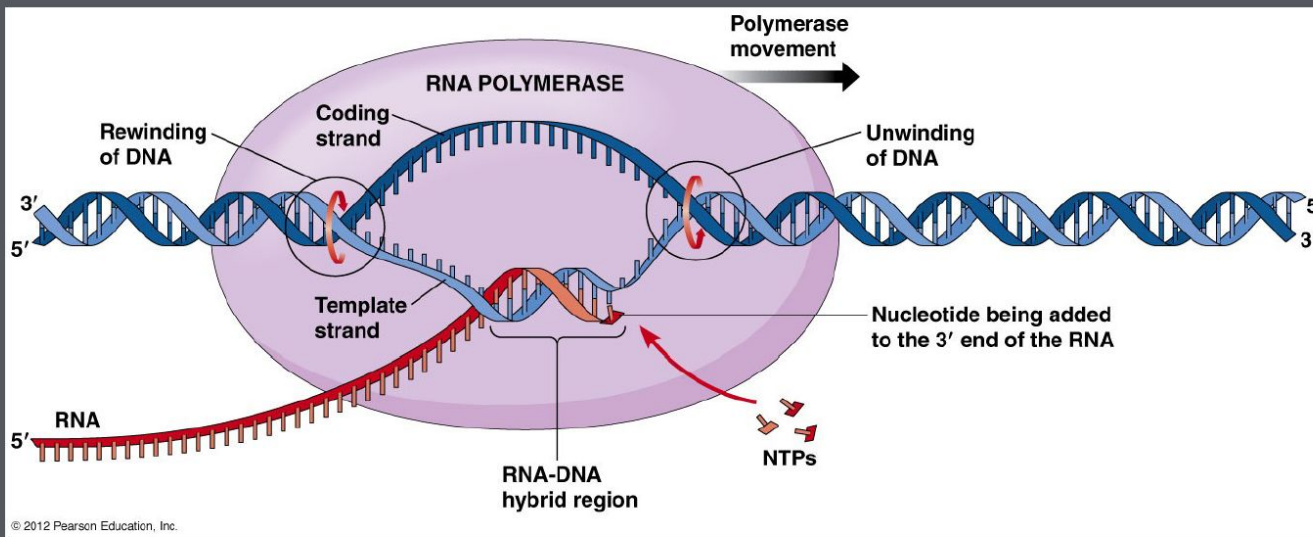
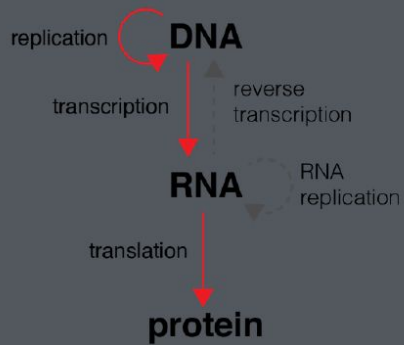


Replication

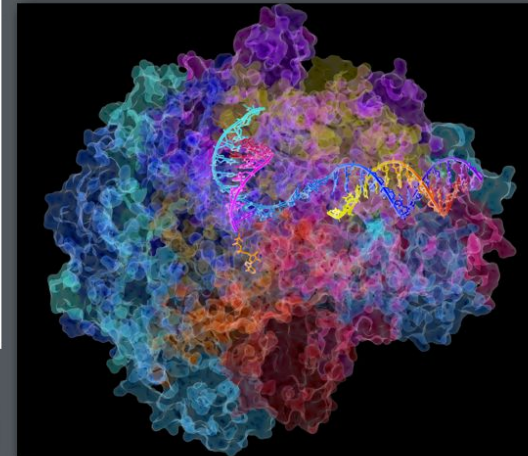


DNA Polymerase

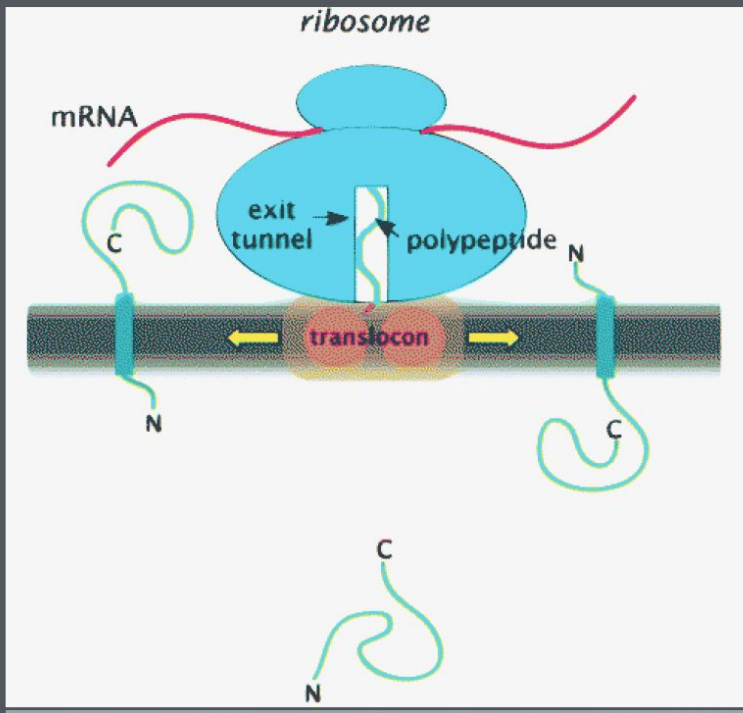
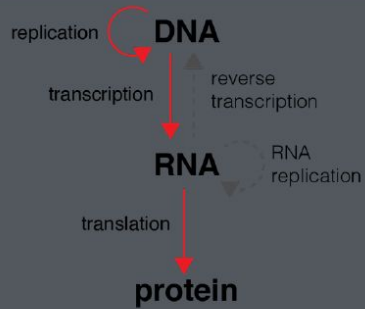
Transcription



Roger Kornberg 2006



Translation



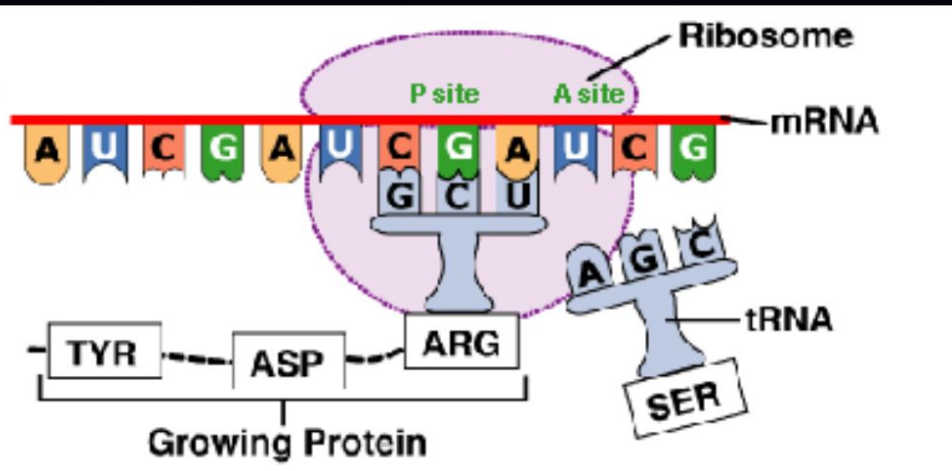
(White & von Heijne, 2005)

Ribosome

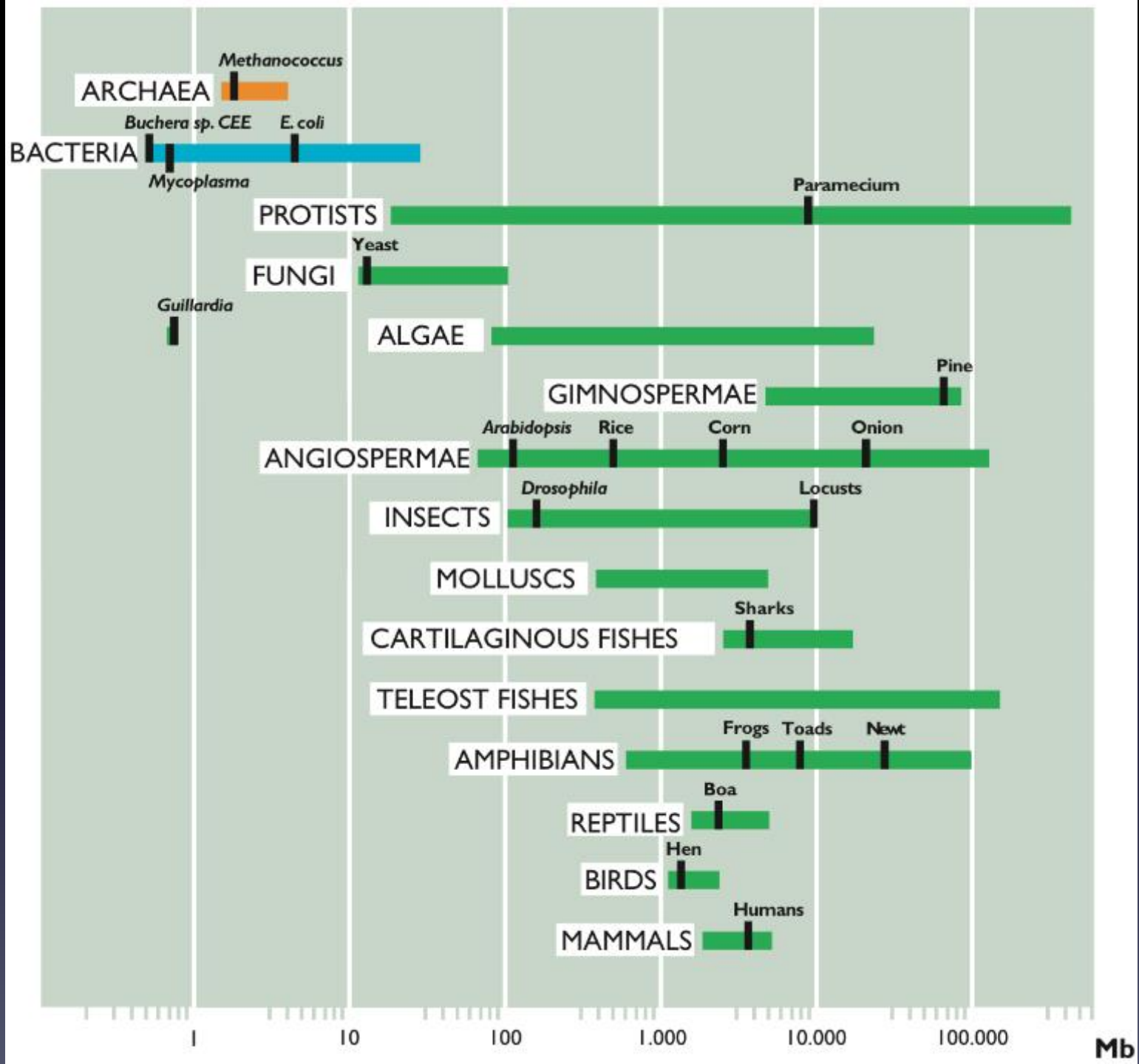


Tom Steitz, Peter Moore,
Venkatraman Ramakrishnan
Nobel 2009

The Genetic Code



		Second Letter				
		T	C	A	G	
First Letter	T	TTT } Phe TTC } TTA } Leu TTG }	TCT } Ser TCC } TCA } TCG }	TAT } Tyr TAC } TAA Stop TAG Stop	TGT } Cys TGC } TGA Stop TGG Trp	T C A G
	C	CTT } Leu CTC } CTA } CTG }	CCT } Pro CCC } CCA } CCG }	CAT } His CAC } CAA Gln CAG }	CGT } Arg CGC } CGA } CGG }	T C A G
	A	ATT } Ile ATC } ATA } ATG Met	ACT } Thr ACC } ACA } ACG }	AAT } Asn AAC } AAA Lys AAG }	AGT } Ser AGC } AGA Arg AGG }	T C A G
	G	GTT } Val GTC } GTA } GTG }	GCT } Ala GCC } GCA } GCG }	GAT } Asp GAC } GAA Glu GAG }	GGT } Gly GGC } GGA } GGG }	T C A G



Key Differences

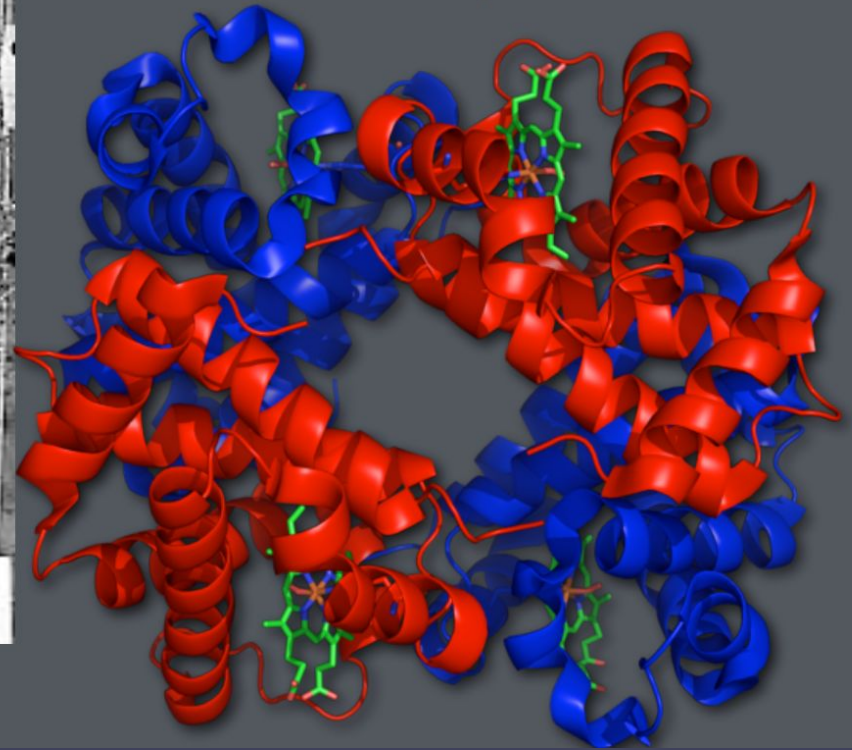
Feature 	Gymnosperms	Angiosperms
Seeds	Naked, exposed on scales (cones).	Enclosed within an ovary, developing into a fruit.
Reproduction	Non-flowering, produce cones (strobili).	Flowering plants; produce true flowers.
Pollination	Primarily wind-pollinated.	Diverse (wind, insects, animals).
Endosperm	Haploid (n), forms before fertilization.	Triploid (3n), formed after double fertilization.
Vascular Tissue	Lack true xylem vessels (except Gnetales).	Possess well-developed xylem with vessels.
Examples	Pines, firs, cycads, ginkgo.	Oaks, roses, grasses, fruit trees (mango, apple).
Diversity	Smaller, older group (e.g., conifers).	Largest, most diverse plant group.

Protein structure

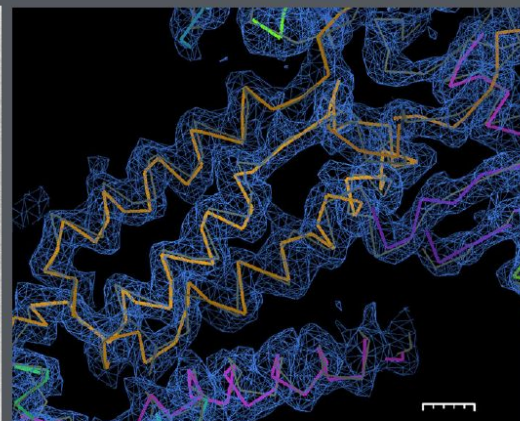
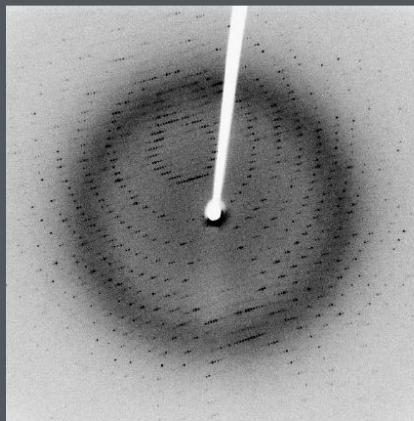
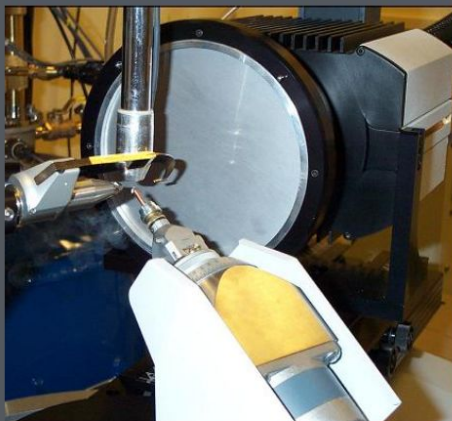
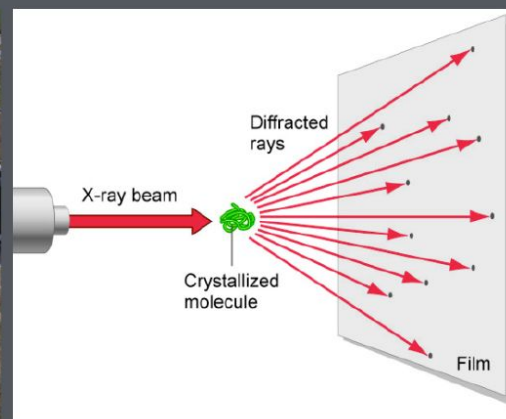


Max Perutz & Hemoglobin

X-ray structure
Took 22 years...



X-ray crystallography



A new hope: Cryo-EM

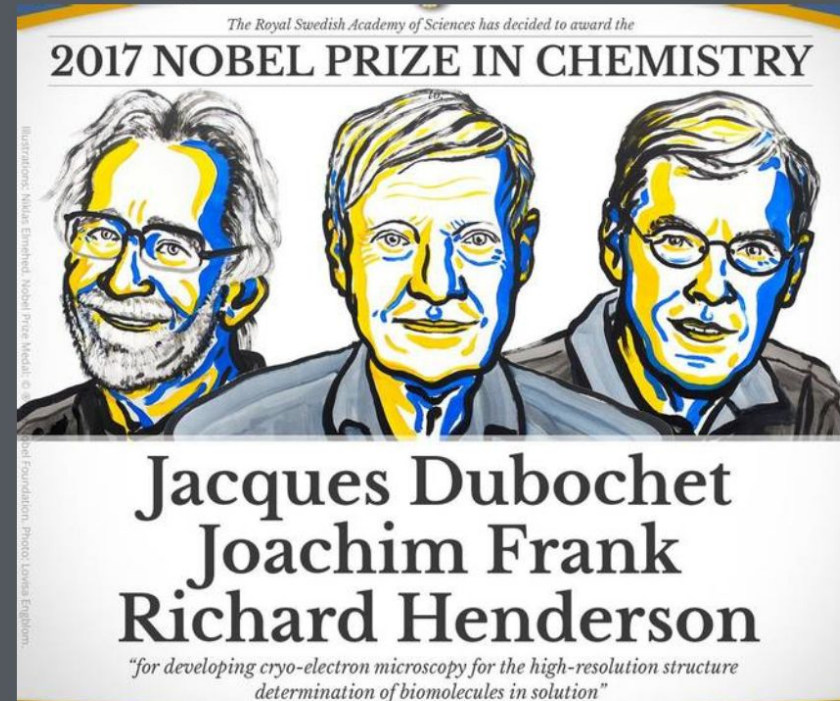


THE REVOLUTION WILL NOT BE CRYSTALLIZED

**MOVE OVER X-RAY CRYSTALLOGRAPHY.
CRYO-ELECTRON MICROSCOPY IS
KICKING UP A STORM IN STRUCTURAL
BIOLOGY BY REVEALING THE HIDDEN
MACHINERY OF THE CELL**

In a basement room, deep in the bowels of a steel-clad building in Cambridge, a major insurgency is under way.

A hulking metal box, some three metres tall, is quietly beaming terabytes' worth of data through thick orange cables that disappear off through the ceiling. It is one of the world's most advanced cryo-electron microscopes: a device that uses electron beams to photograph frozen biological molecules and lay bare their molecular shapes. The microscope is so sensitive that a shout can ruin an experiment, says Sjors Scheres, a structural biologist at the UK Medical Research Council Laboratory of Molecular Biology (LMB), as he stands dwarfed beside the £5-million (US\$7.7-million) piece of equipment. "The UK needs many more of these. Some really think it's time to be a power. The reality is



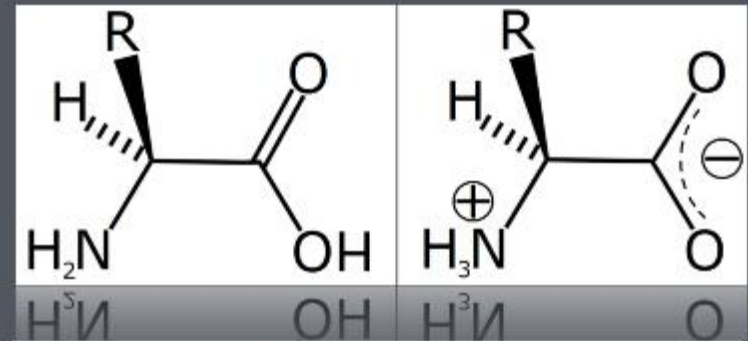
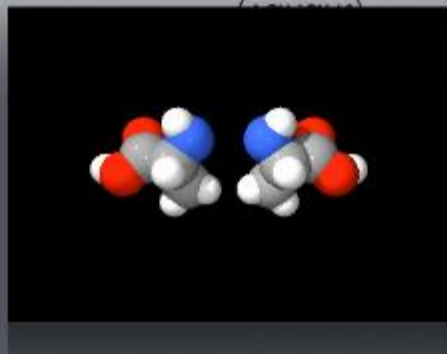
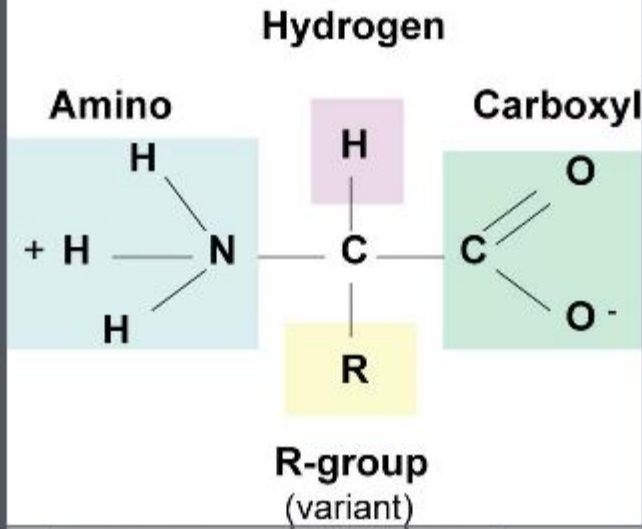
Nature
September 2015

Study questions

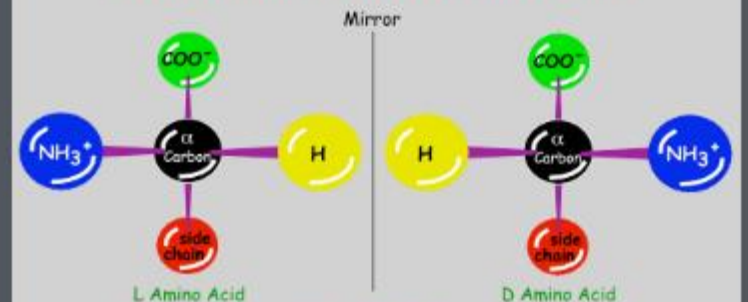
1. What are the time & length scales involved in biophysics? What do they represent? Why is water such a special molecule in biophysics?
2. What is the size of a protein?
3. What are hydrogen bond donors and acceptors?
4. What is the energy of a hydrogen bond? In water? In a protein?
5. What are the (structural) components of nucleic acids?
6. Name three pieces of information enabling DNA structure determination
7. How many **bases** are there in the human genome?
8. How many **proteins** does the human genome code for?
9. How is the DNA double helix stabilized?
10. What does complementarity (in DNA) mean?
11. How does RNA differ structurally from DNA?
12. How can RNA perform biological function?

Amino acids

Amino Acid Structure



Common Amino Acids are Stereoisomers,
Meaning they have a Chiral α Carbon center.



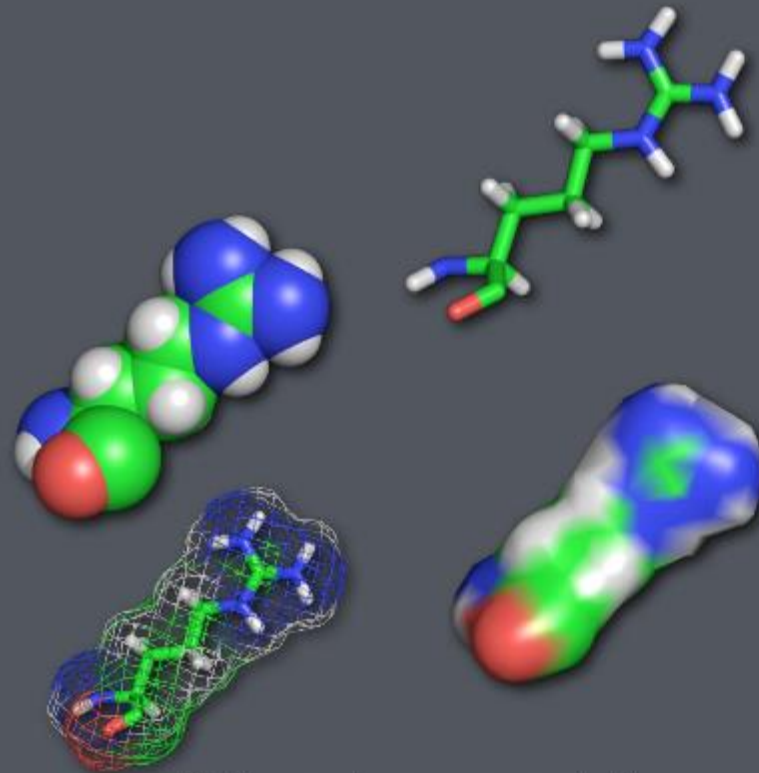
Amino Acids can exist in either the D or L configuration.
However, All Chiral Amino Acids in Proteins have the L configuration

However, All Chiral Amino Acids in Proteins have the L configuration

D/L forms will not convert spontaneously (racemization)

Amino acids

NONPOLAR, HYDROPHOBIC		POLAR, UNCHARGED	
	R GROUPS		
Alanine Ala A MW = 89	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}_3 \end{array}$	$\begin{array}{c} \text{H} - \text{CH} - \text{COO}^- \\ \\ \text{N}^+ \text{H}_3 \end{array}$	Glycine Gly G MW = 75
Valine Val V MW = 117	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}(\text{CH}_3)_2 \end{array}$	$\begin{array}{c} \text{HO} - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{N}^+ \text{H}_3 \end{array}$	Serine Ser S MW = 105
Leucine Leu L MW = 131	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}_2 - \text{CH}(\text{CH}_3)_2 \end{array}$	$\begin{array}{c} \text{OH} - \text{CH} - \text{CH} - \text{COO}^- \\ \quad \\ \text{CH}_3 \quad \text{N}^+ \text{H}_3 \end{array}$	Threonine Thr T MW = 119
Isoleucine Ile I MW = 131	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}(\text{CH}_3) - \text{CH}_2 - \text{CH}_3 \end{array}$	$\begin{array}{c} \text{HS} - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{N}^+ \text{H}_3 \end{array}$	Cysteine Cys C MW = 121
Phenylalanine Phe F MW = 131	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}_2 - \text{C}_6\text{H}_5 \end{array}$	$\begin{array}{c} \text{HO} - \text{C}_6\text{H}_4 - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{N}^+ \text{H}_3 \end{array}$	Tyrosine Tyr Y MW = 181
Tryptophan Trp W MW = 204	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}_2 - \text{C}_8\text{H}_6\text{N}_2 \end{array}$	$\begin{array}{c} \text{NH}_2 - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{N}^+ \text{H}_3 \end{array}$	Asparagine Asn N MW = 132
Methionine Met M MW = 149	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{S} - \text{CH}_3 \end{array}$	$\begin{array}{c} \text{NH}_2 - \text{C}(=\text{O}) - \text{CH}_2 - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \\ \text{N}^+ \text{H}_3 \end{array}$	Glutamine Gln Q MW = 146
Proline Pro P MW = 115	$\begin{array}{c} \text{OOC}^- \\ \\ \text{CH} - \text{CH}_2 - \text{CH}_2 \\ \quad \\ \text{HN} - \text{CH}_2 \end{array}$	POLAR BASIC $\begin{array}{c} \text{NH}_3^+ - \text{CH}_2 - (\text{CH}_2)_3 - \text{CH} - \text{COO}^- \\ \\ \text{N}^+ \text{H}_3 \end{array}$	Lysine Lys K MW = 146
Aspartic acid Asp D MW = 133	POLAR ACIDIC $\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}_2 - \text{C}(=\text{O})\text{O}^- \end{array}$	$\begin{array}{c} \text{NH}_2 - \text{C}(=\text{O}) - \text{NH} - (\text{CH}_2)_3 - \text{CH} - \text{COO}^- \\ \quad \\ \text{N}^+ \text{H}_3 \quad \text{N}^+ \text{H}_3 \end{array}$	Arginine Arg R MW = 174
Glutamic acid Glu E MW = 147	$\begin{array}{c} \text{OOC}^- \\ \\ \text{H}_2\text{N}^+ - \text{CH} - \text{CH}_2 - \text{CH}_2 - \text{C}(=\text{O})\text{O}^- \end{array}$	$\begin{array}{c} \text{HN} = \text{C} - \text{CH}_2 - \text{CH} - \text{COO}^- \\ \quad \\ \text{NH}_2 \quad \text{N}^+ \text{H}_3 \end{array}$	Histidine His H MW = 155

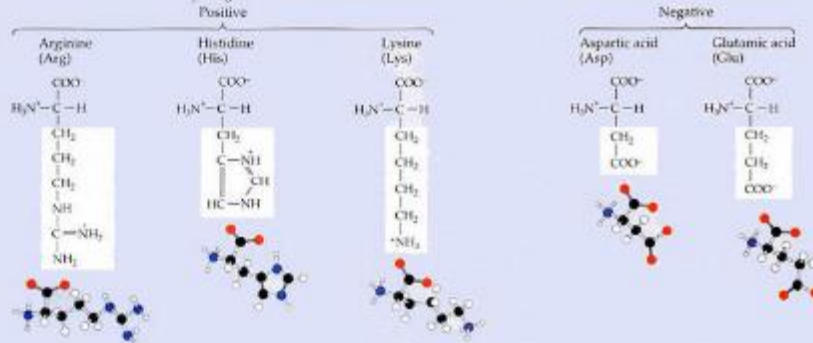


Different representations of arginine amino acid

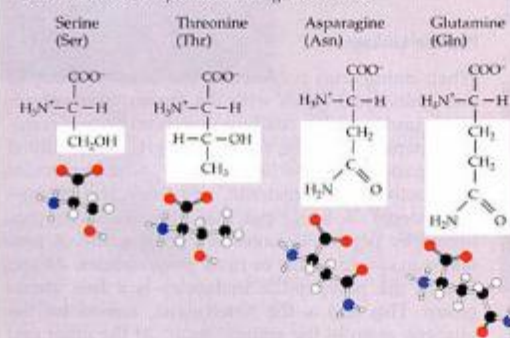
Natural amino acids

Learn how to classify them (hydrophobic, hydrophilic, charge)

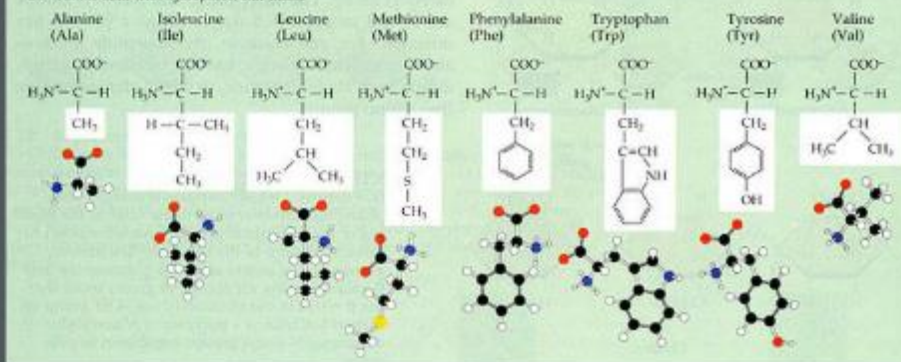
A. Amino acids with electrically charged side chains



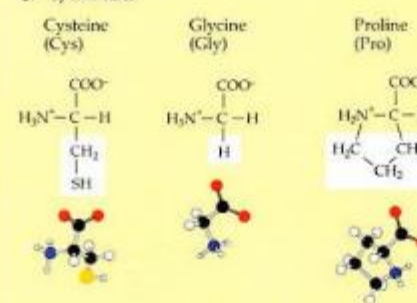
B. Amino acids with polar but uncharged side chains



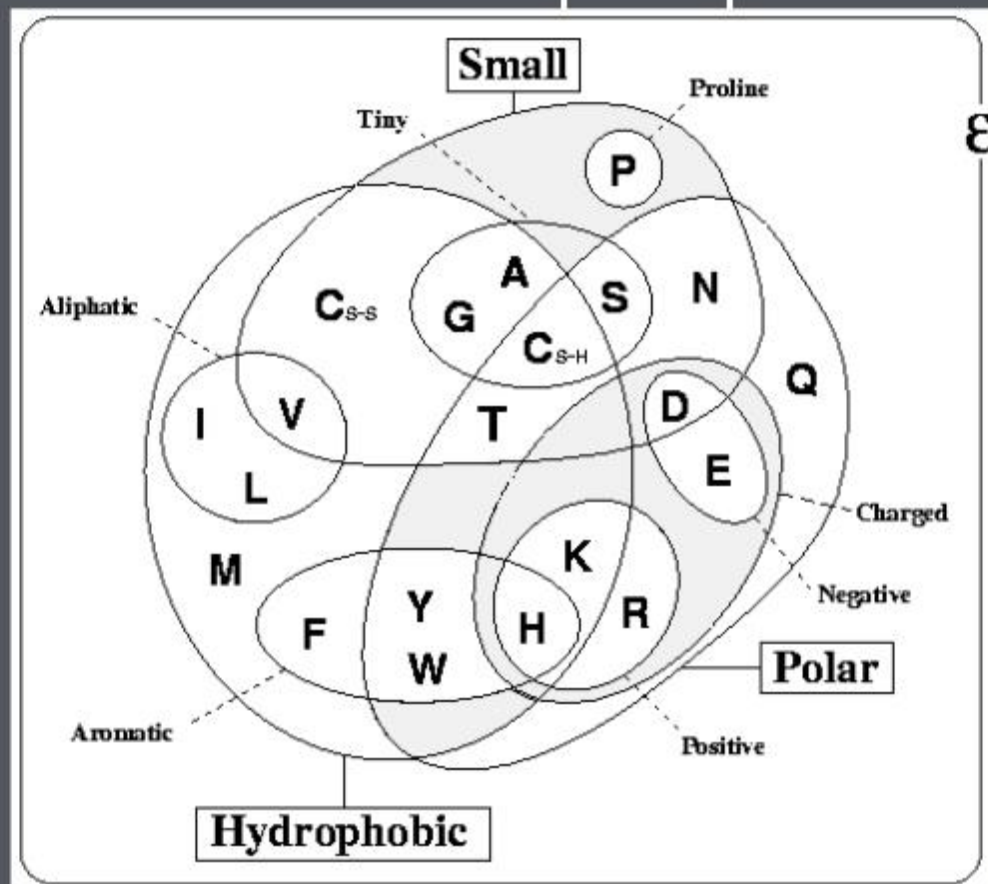
D. Amino acids with hydrophobic side chains



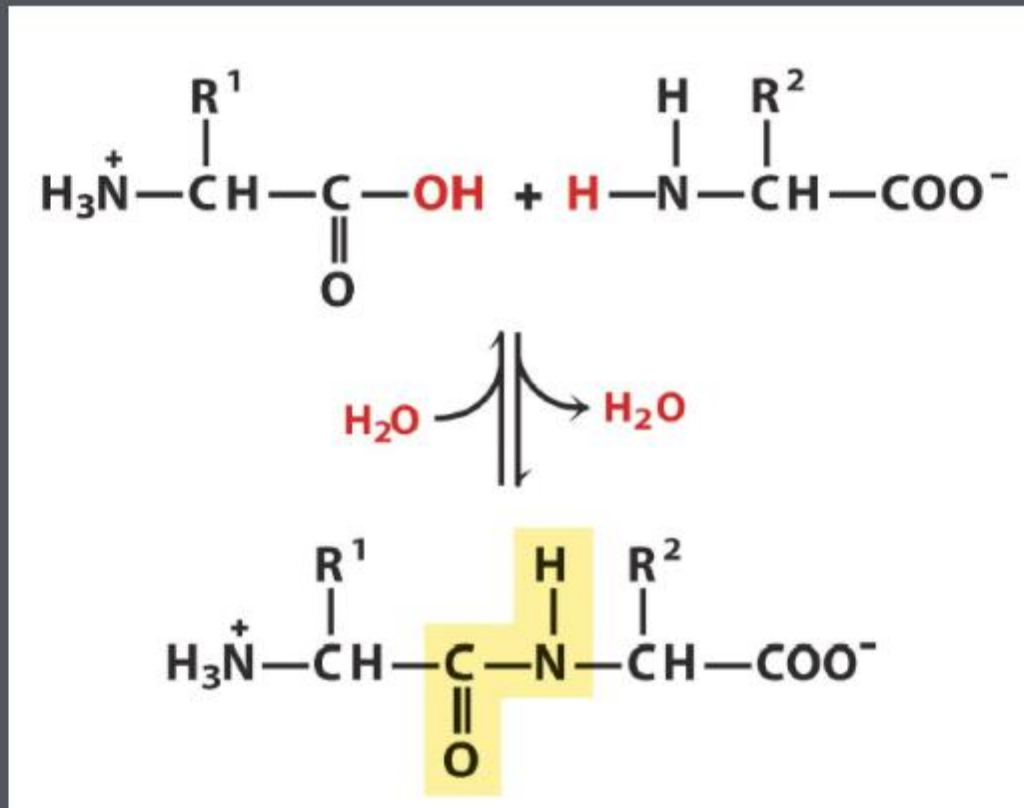
C. Special cases



Amino acid properties

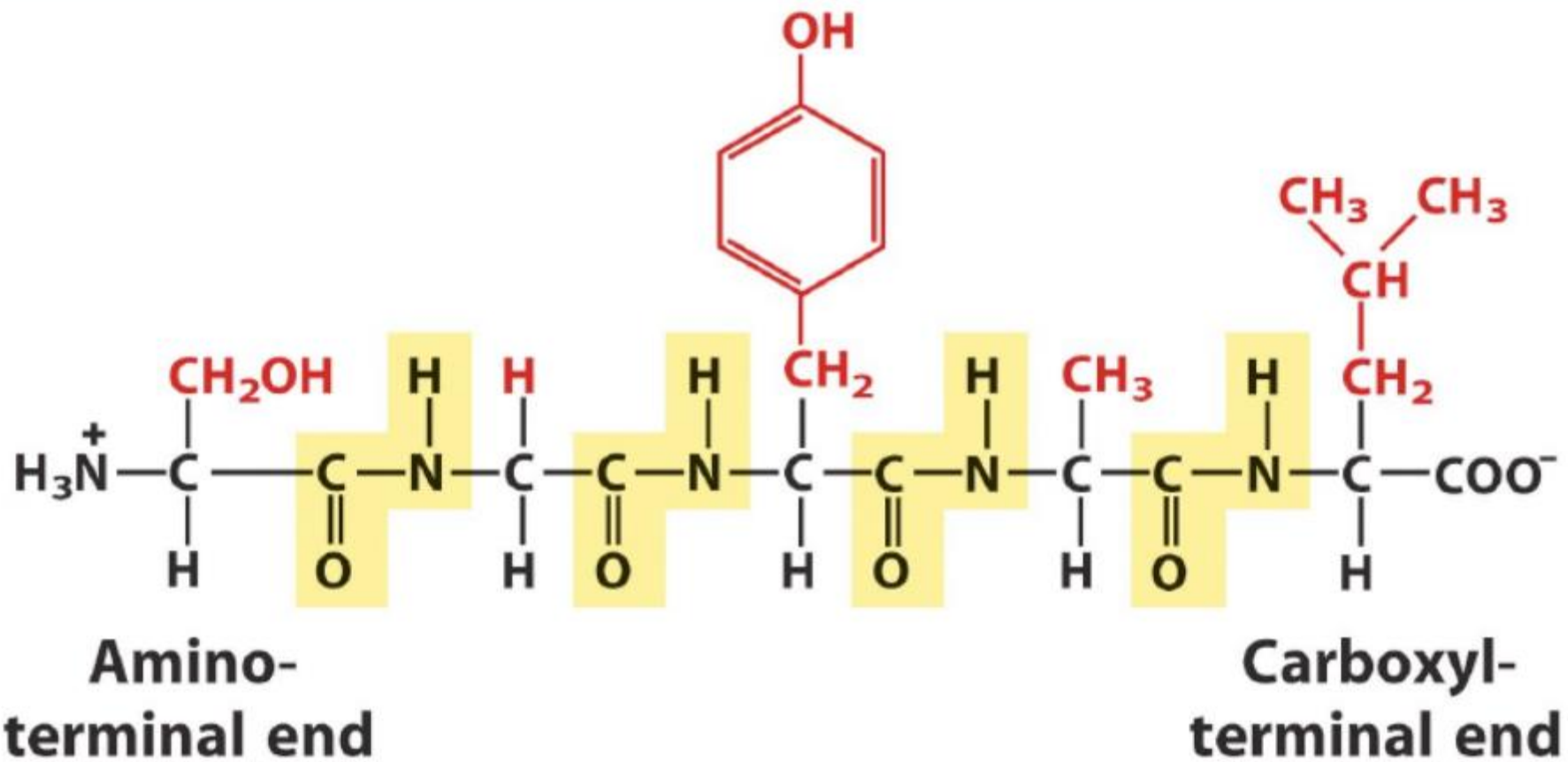


Polymerization



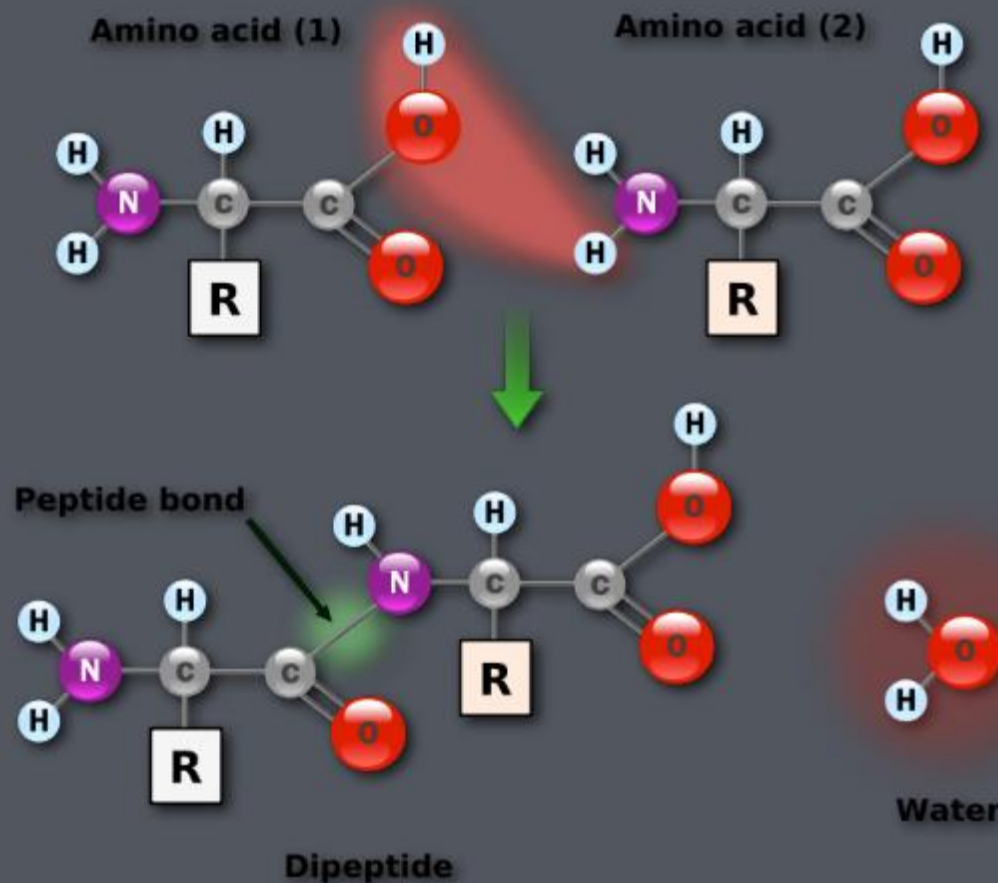
Atoms form bonds to make their electrons more stable
A bond corresponds to delocalizing electrons

Polypeptides



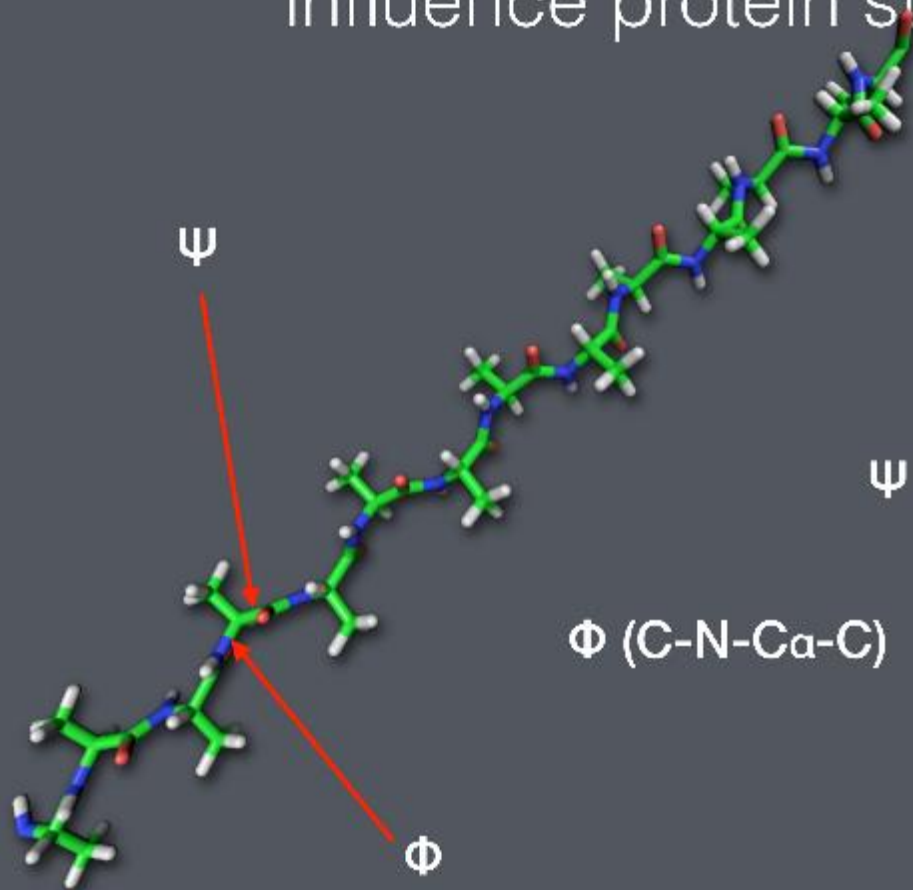
Peptide bonds

Peptide bond
will be planar
due to electron
resonance;
sp²-hybridization
of N & C atoms



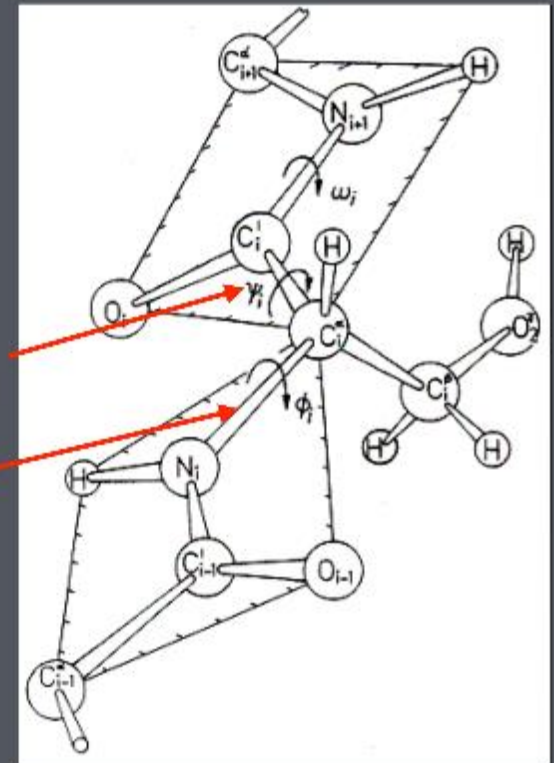
It is also rigid
due to extra
electrons
("double bond")

Discussion: What motion(s) influence protein structure and why?



Ψ (N-C α -C-N)

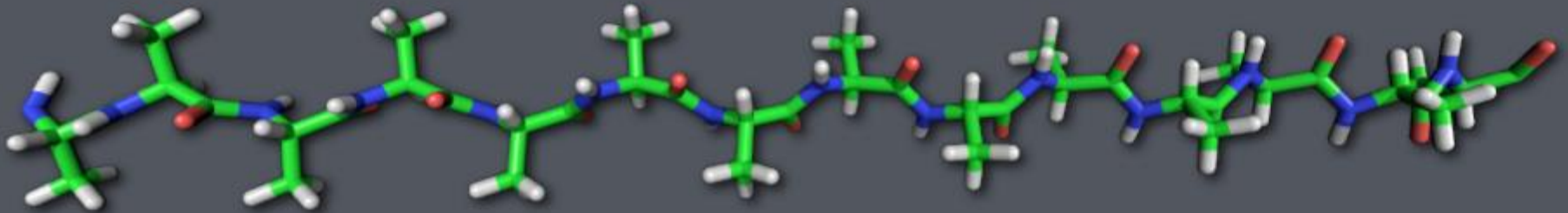
Φ (C-N-C α -C)



Polypeptide structure

- Backbone degrees of freedom:
 - Peptide (Ω) bond (trans (cis only for Pro))
 - Φ (C-N-CA-C), Ψ (N-CA-C-N) torsions
- Side chain degrees of freedom:
 - χ_1, χ_2, χ_3 torsions

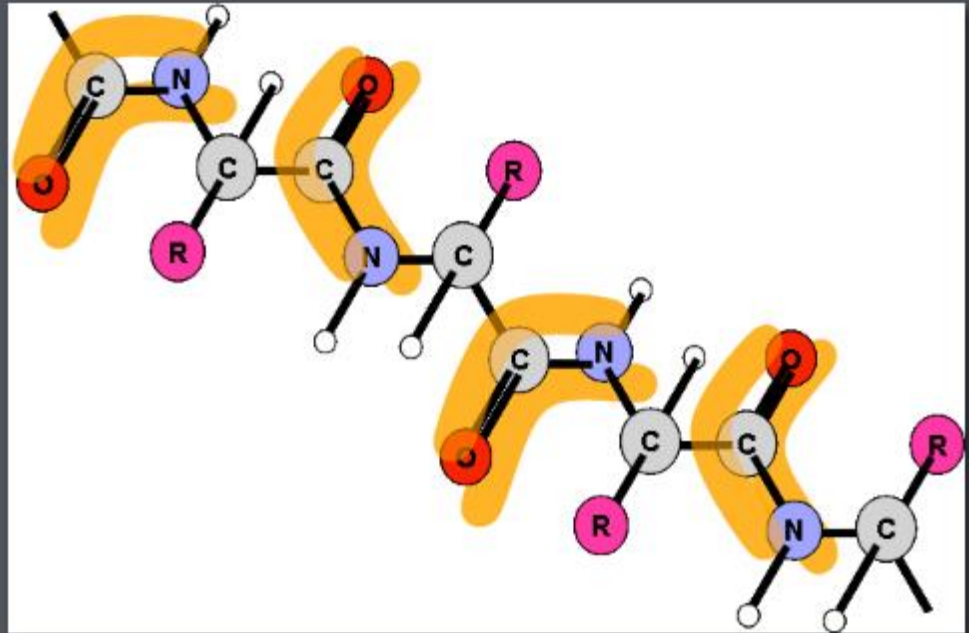
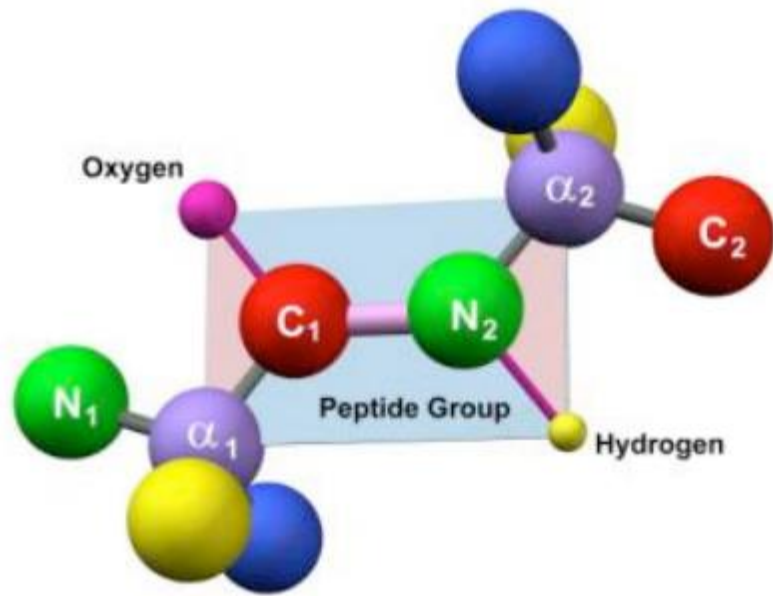
Rotational barrier around
"single" ($\text{CH}_3\text{-CH}_3$) bond
is roughly 2-4 kcal/mol



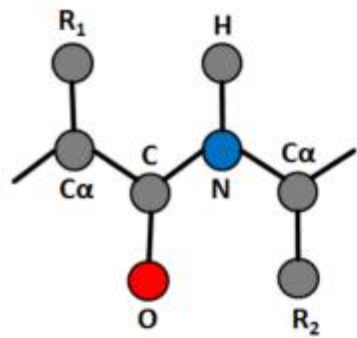
Conformational space

- How many conformations are there?
- Sample Φ, Ψ torsion in 10 degree units
 - 36 'states' for each torsion
- For a 100-residue chain we get:
 - 36^2 states per residue
 - $(36^2)^{100} = 36^{200} \approx 10^{308}$ states for the chain
- Only one is the native structure

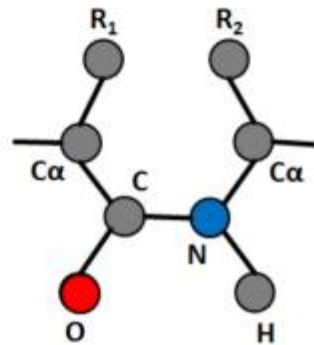
Cis/trans isomerization



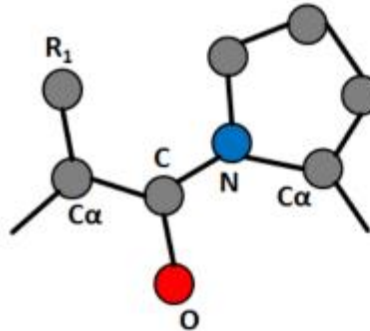
Cis/trans isomerization



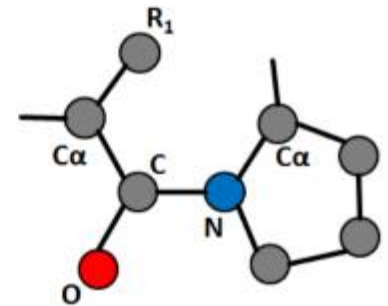
trans



cis



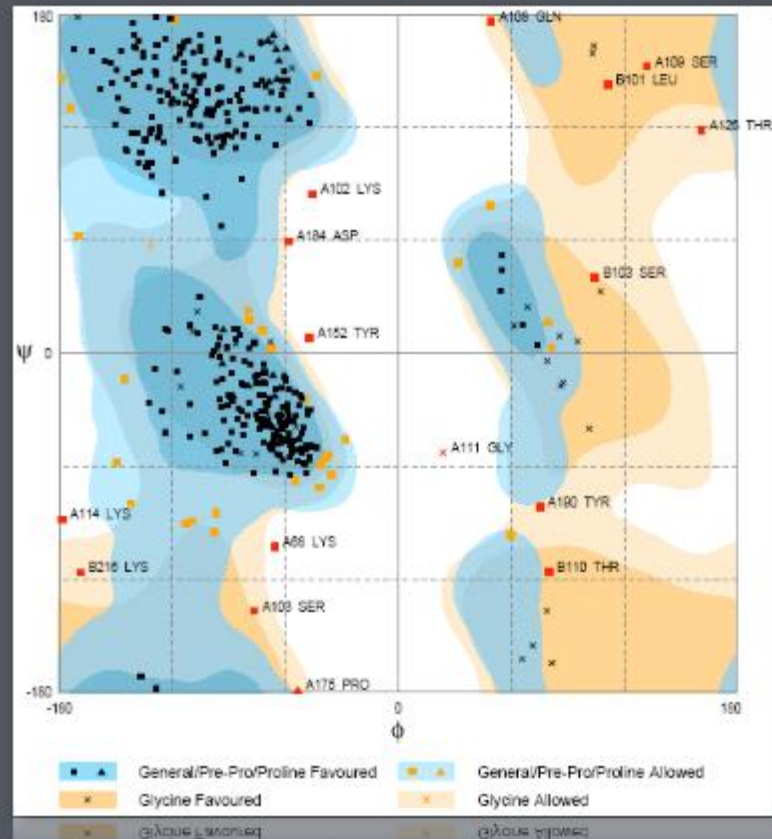
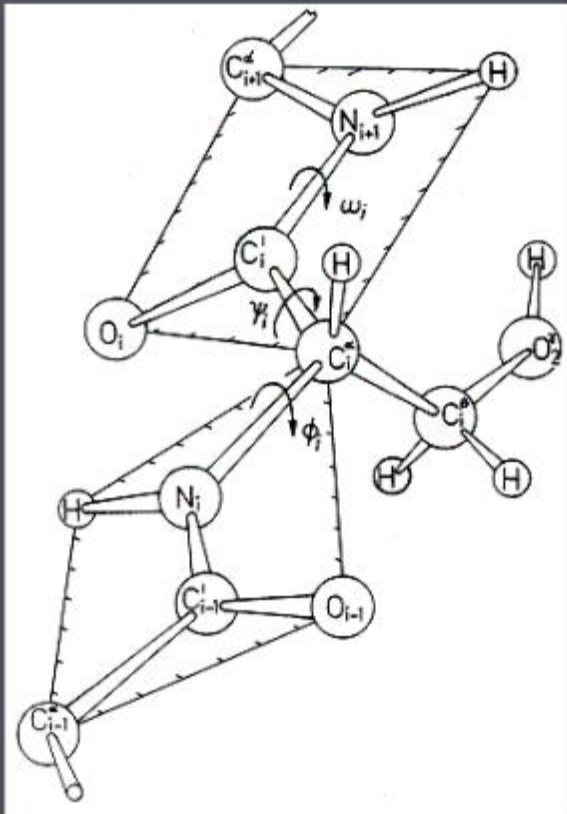
cis



trans

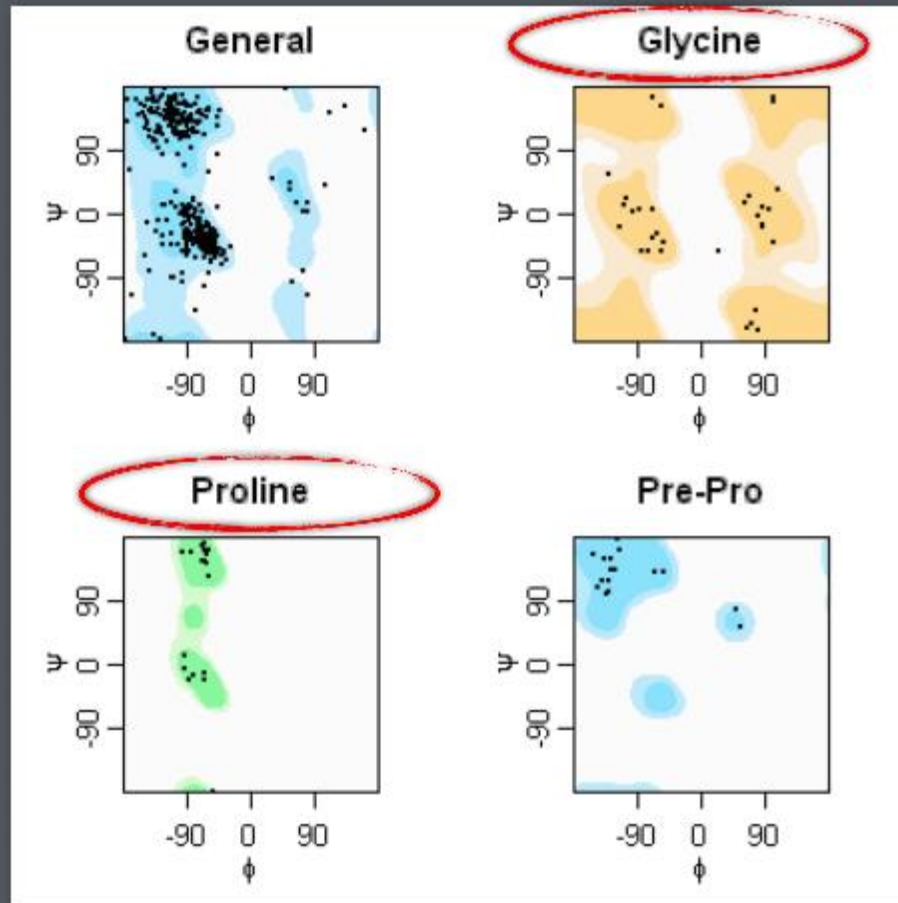
Proline is a special case!

Ramachandran diagrams

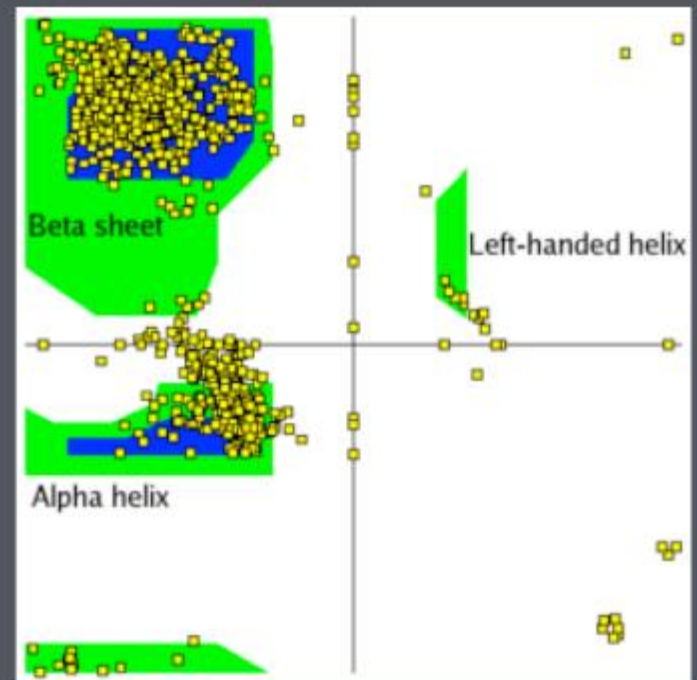


Natural degrees of freedom for amino acids

Ramachandran species

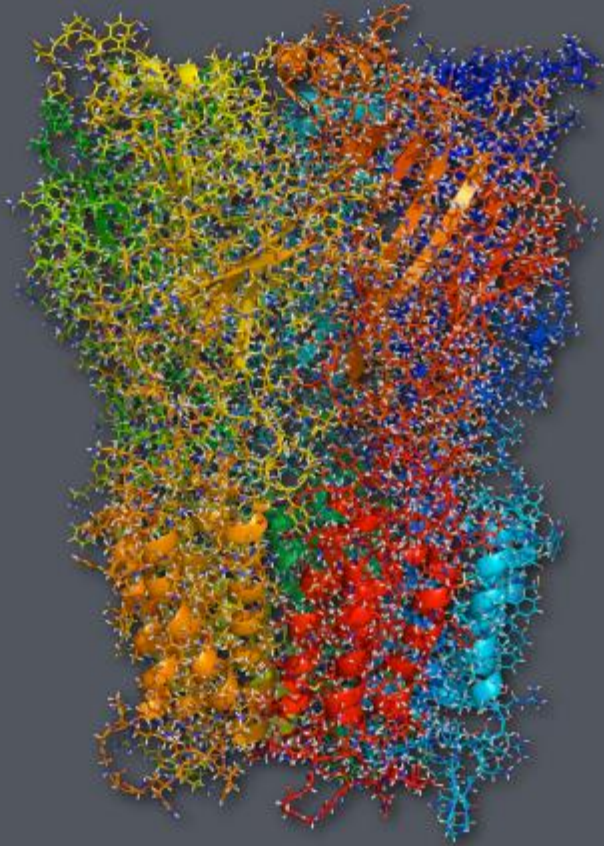


Glycine very flexible (no side chain)
Proline very inflexible; even messes up previous residue!

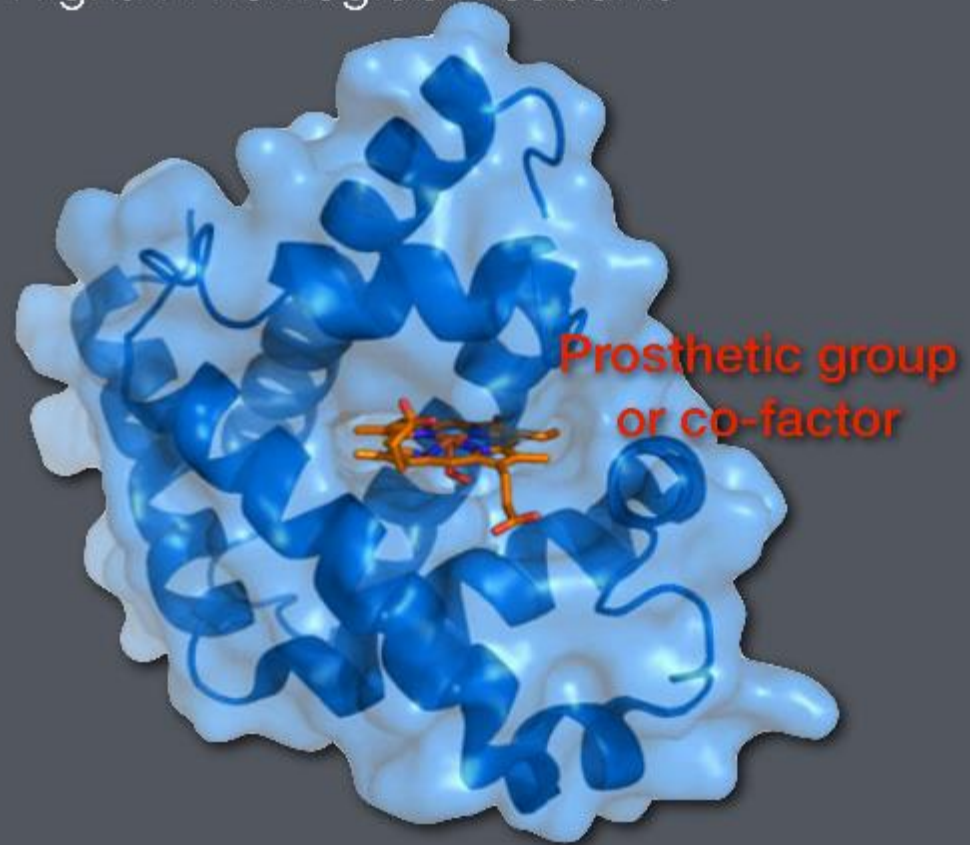


An assembled protein

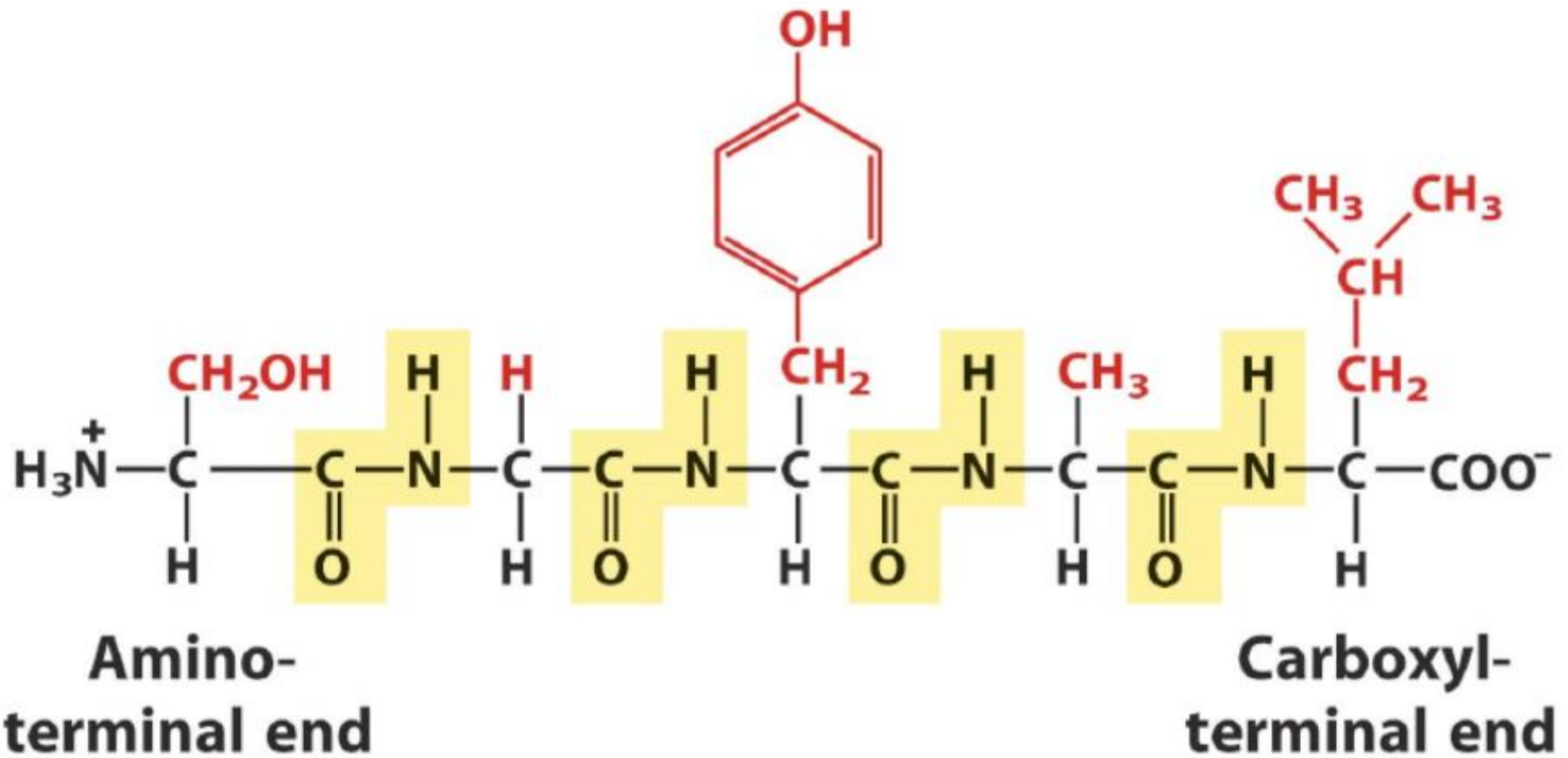
Left: The GLIC ion channel



Right: A hemoglobin subunit



Why this diversity?



Anfinsen & Levinthal

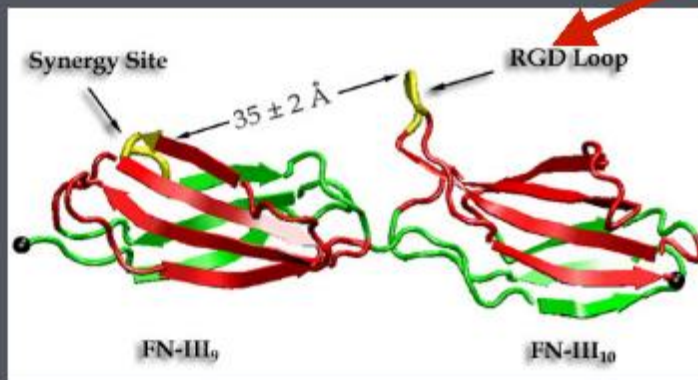
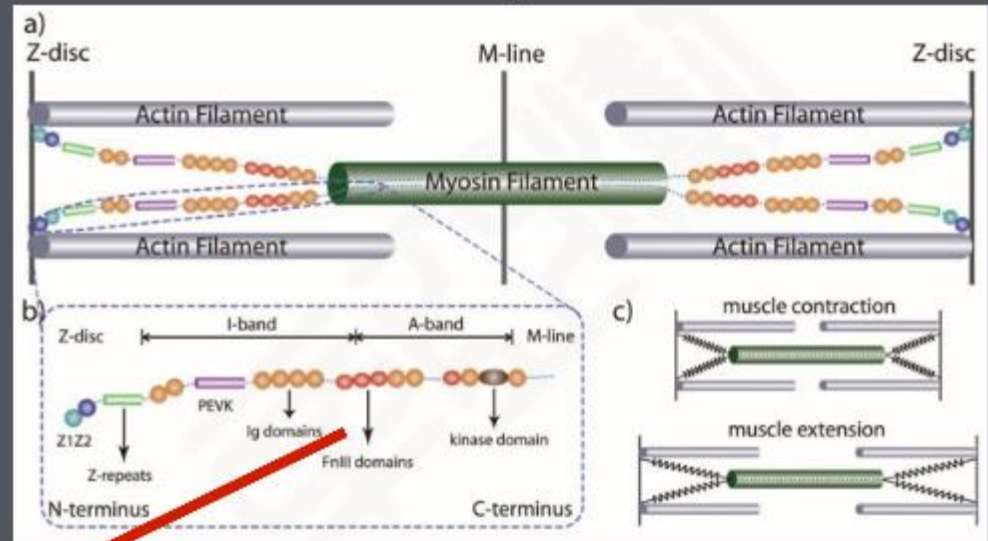
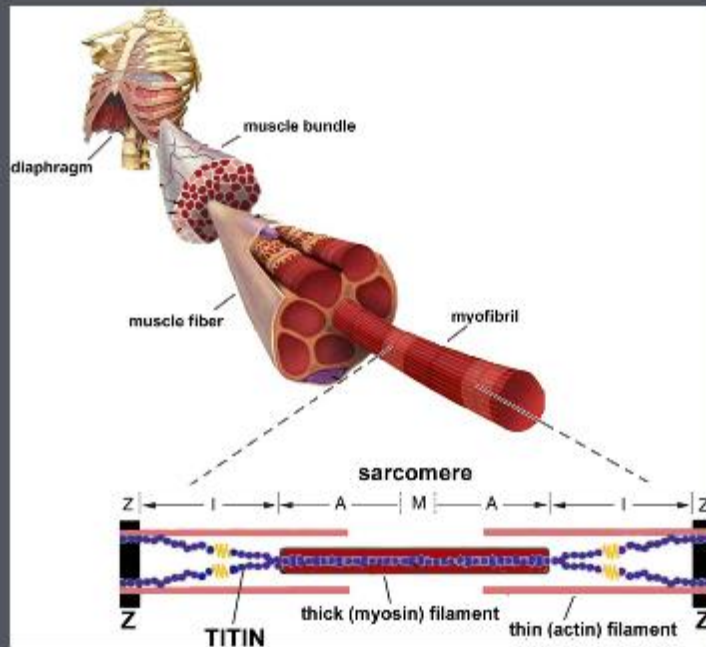
- Christian Anfinsen: Proteins adopt the structure corresponding to the global minimum in free energy (Nobel 1972)
- Cyrus Levinthal: But how can a protein find that? Even with just 2 states per amino acid and 100 residues there are 2^{100} states! (*Levinthal's Paradox*)



Example Proteins

	<i>Molecular weight</i>	<i>Number of residues</i>	<i>Number of polypeptide chains</i>
Cytochrome c (human)	13,000	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (chicken egg white)	13,930	129	1
Myoglobin (equine heart)	16,890	153	1
Chymotrypsin (bovine pancreas)	21,600	241	3
Chymotrypsinogen (bovine)	22,000	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	68,500	609	1
Hexokinase (yeast)	102,000	972	2
RNA polymerase (<i>E. coli</i>)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (<i>E. coli</i>)	619,000	5,628	12
Titin (human)	2,993,000	26,926	1

Case study: Titin



Covered material and suggested reading

- Finkelstein
 - Chapter 3 & 4
- David Eisenberg - "The discovery of helices and sheets, the principal structural features of proteins", PNAS, 2003
- Cyrus Levinthal - "Molecular Model building by Computer", Scientific American, 1966

Study questions

1. What are the time & length scales involved in biophysics? What do they represent? Why is water such a special molecule in biophysics?
2. What is the size of a protein?
3. What are hydrogen bond donors and acceptors?
4. What is the energy of a hydrogen bond? In water? In a protein?
5. What are the (structural) components of nucleic acids?
6. Name three pieces of information enabling DNA structure determination
7. How many **bases** are there in the human genome?
8. How many **proteins** does the human genome code for?
9. How is the DNA double helix stabilized?
10. What does complementarity (in DNA) mean?
11. How does RNA differ structurally from DNA?
12. How can RNA perform biological function?

Study questions

1. Name three properties by which amino acids can differ.
2. One amino acid is NOT chiral. Which one? Is it special in any way?
3. There is one more highly special amino acid. Which one? How is it special?
4. What would happen if we had some D-isomer amino acids in a protein?
5. How are amino acids linked into a protein chain?
6. What are the levels of structure organization in proteins?
7. Describe the relation between sequence, structure, and function.
8. Does function induce structure?
9. Which ones are the most important degrees of freedom in proteins?
10. Why are cis conformations almost never observed in peptide bonds?
11. What did (1) Anfinsen and (2) Levinthal say?
12. Explain what a Ramachandran plot can tell you about protein geometry.